UNIT 1 INTRODUCTION TO INTERNET

1.0 INTRODUCTION

Today it is the era of computers, to be more specific, the digital data. Every organization, small or big in size, doing business from a single site or multiplesite, from one nation or over the seas, not only uses computers to handle various transactions but also communicate information among various sites located over various geographically distant places. The means by which the information is exchanged among computers separated apart is known as the computer Internet. a network is; entities connected to each other to exchange some information. when entities are replaced by computing devices, it is the computer network. Digital information is made available in just one click from one end of the world to another end within fraction of seconds. The productivity of every organization irrespective of their field has been increased enormously by sharing of information in digital form through computer network. A Computer network is two or more standalone computers connected to each other to share information, resources, messages etc irrespective of their installed geographical location.

Further, in section 1.1 the objectives of this unit are listed, one should be able to achieve these after completion of this unit. In section 1.2 briefly discussed about the Internet. Network taxonomy, public and private networks and how Internet can be accessed are discussed in brief in this section. Section 1.3 discusses about the various network topologies.lNetwork Models are discussed in the section 1.4. The two network models namely: OSI model and TCP/IP model are discussed here. Section 1.5 summerizes the unit. In section 1.6 solution and their answers are covered for the unit.

1.1 OBJECTIVES

After completing this unit, one should be able to:

- understand the term computer networks;
- differentiate between different types of computer networks;
- identify the advantages of using computer networks.
- identify the applications of computer networks;
- understand various network topologies;
- know the significance of network protocols;
- understand the two computer network models: OSI and TCP/IP reference model;

1.2 WHAT IS THE INTERNET?

The Internet is the biggest computer network with a world-wide scope. It is the network of networks. The Internet comprises lakhs of organizational networks of types: domestic orinternational, academic or business, and government, which altogether exchange various kinds of data/information. In routine we generally use the term 'net' for Internet. It connects billions of people through computers or mobiles from every corner of the world to share information. Worlds first general purpose computer network was ARPANET. ARPANET. The Bob Taylor is known as the father of the Internet. He initiated the project in 1966 to enable access to remote computers. As a result, in 1969 first time in a computer is accessed remotely. This was the start of what we see today around us in digital information and computer world.

The public Internet is a world-wide computer network i.e. millions of computing devices are connected to each other spread through out the world. Computing devices may be desktop computers, laptops, mobile phone, Web TVs, installed with various kind of Operating Systems i.e. windows based, android based, linux/unix based mac-os based etc. In the Internet jargon, these devices are called hosts or end systems. End systems use network applications such as WWW (browsers) and e-mail etc to connect with Internet.

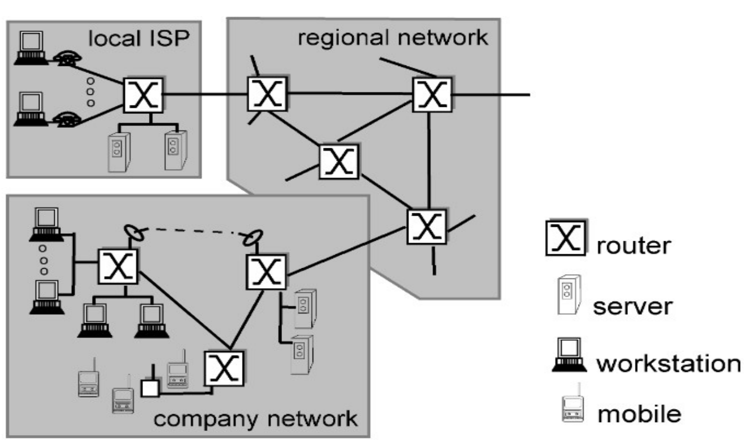

Figure1 : Overview of the Internet

1.2.1 ISP and Internet Backbone

ISP

ISPs (Internet Service Provides) are responsible for transporting of traffic in the Internet. Internet is very vast in terms of traffic, number of end systems, number of intermediate devices, interconnecting links and distribution of these over geographical locations. Hence huge number of ISPs manage the Internet. To efficiently manage and control the services of Internet, these ISPs are classified intothree categories namely: tier -1 ISPs, tier -2 ISPs and tier -3 ISPs and this hierarchical model is known as 3-tiermodel. An ISP is categorized as tier -1, 2 or 3 based on the type of Internet services they provide.

Any one of us can become an access ISP as soon as we have an Internet connection. All we need to do is buy the required hardware/software (ex. router, modem, firewall etc) to allow other users/customers to connect to us. Adding new tiers and branches to the Internetis as simple as adding a new piece of Lego to an existing Lego construction.

Internet Backbone

Internet backbone is named for core network, can handle traffic at very high speed, interconnecting various networks to allow transmission of data over the worldwide network. Internet backbone mainly consists of tier-1 ISP's networks altogether. It connects networks of all tier -1 ISP and carries the majority of Internet traffic. Internet backbone mainly contains a large number of routers, switches and other intermediate devices interconnected with very high speed links (optical fiber links). Many companies may build their own backbone infrastructure for interconnectivity among offices at different geographical locations. Similar is the case with Internet backbones, where the computer connects to an access network of a tier -3 ISP, which further connects to the main Internet backbone (tier -2 or tier -1 ISP's network)to access the worldwide network.

1.2.2 Interconnection of ISPs

End systems or hosts situated at the edge of the Internet are connected to the rest of the Internet through this 3-tier model of ISPs, as shown in Figure. The network through wich end systems connect to the Internet is called as the access network (i.e. local LAN or private network). Access network connects to the Internet via tier -3 ISPs also known as access ISPs and are at the bottom of 3 –tier model. Access ISPs are the largest number of ISPs in the Internet. Tier -1 ISPs are relatively less in numbers and are at the top of the hierarchy. All three tiers of ISPs are same in many ways as any network, like they all have connecting links and intermediate devices i.e. routers and are further connected to other networks/ISPs.

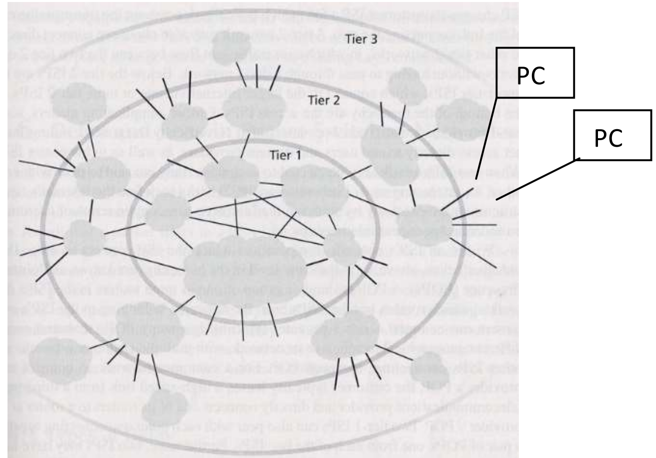

Figure 2 : tier model of ISPs

On the other hand, each of the tiers of ISPs are special and responsible to perform specific jobs. Tier-1 ISPs have links of speed of the range 622 Mbps to 2.5 to 10 Gbps. The routers of tier-1 ISPs must be able to handle traffic in accordance with extremely high rate links. Tier-1 ISPs are directly connected to each of the other tier-1 ISPs. The coverage of tier -1 ISPs is the entire Internet spread to all over the world.

Tier -1 ISPs are also known as Internet backbone networks. Some of the tier -1 ISP are Sprint, MCI (previously UUNet/WorldCom), AT&T, Level3 (which acquired Genuity), Qwest, and Cable & Wireless. Interestingly, no group officially sanctions tier-1 status; as the saying goes—if you have to ask if you are a member of a group, you're probably not. Tier -1 ISPs are connected to a large number of tier-2 ISPs.

Tier-2 ISPs have regional or national coverage, and are connected to only a few of the tier-1 ISPs. Thus, tier -2 ISPs need to route traffic through one of the teir -1 ISPs to reach a large portion of the global Internet. A tier-2 ISP is a customer of the tier-1 ISPs it is connected with, and teir -1 ISPs are the service providers for their connected customers.

Companies and institutions requiring high speed and less delay connect their network directly to a tier-1 or tier-2 ISPs. Service provider ISP charges its customer ISP a fee, depending on the transmission rate of the link connecting both. A tier-2 ISP network may choose to connect directly with other tier-2 ISP’s networks. When two tier -2 ISPs are directly connected with each other, the traffic of these networks can flow directly between them without having to pass through a tier-1 network. Below the tier-2, are the tier-3 ISPs, which connect to the larger Internet via one or more tier-2 ISPs. At the lowest level of the hierarchy are the access ISPs. End systems or hosts are connected to Internet via access ISPs. When two ISPs are directly connected to each other, they are said to be peer ISPs.

The points (router) at which an ISP connect to other ISPs (of tier above, below or of same tier) are known as Points of Presence (POPs). A tier-1 provider has many POPs in its network spread over various geographical locations, where other ISP’s routers and customer networks are connected. A customer leases a high-speed link to connect its one of the router to a router in a provider's network POP. Furthermore, two ISPs may have multiple peering points connecting with each other at two or more pairs of POPs.

In summary, the topology of the Internet is complex, consisting of dozens of tier-1 and tier-2 ISPs and thousands of lower-tier ISPs. The ISPs are diverse in their coverage, with some spanning multiple continents and oceans, and others limited to narrow regions of the world. The lower-tier ISPs connect to the higher-tier ISPs. Users and content providers are customers of lower-tier ISPs, and lower-tier ISPs are customers of higher-tier ISPs.

1.2.2.1 Taxonomy of Network

Computer networks are generally classified according to their structure, physical scales, and transmission technology etc. According to the network's physical area, it can be classified as:

Personal Area Network (PAN): PAN is for a person. It is a network established for a person for a small area of the range of few metere. It can be wired (using USB data cable) or wireless (using infrared, ZigBee, Bluetooth). To establish PAN no third party device is required, the devices connect to each other autonomously detects and connect with each other. Examples of wireless PAN, or WPAN are: controlling TV with remote using infrared, connecting mobile headsets with mobile using Bluetooth, wireless keyboards and mice, wireless printers , wireless CCTV cameras etcusing Wi-Fi, connecting printer, scanner, bar code scanners and game consoles using USB connection.

• **Local Area Network (LAN):**The physical scope of a Local area network is limited to a small geographical area, such as single building or a campus. Within an organization or institution personal computers are connected among themselves and establish a local area network to share resources i.e. printers , scanners etc and to share information.

• **Metropolitan Area Network (MAN):**The coverage of a MAN is spread over a city and limited to 100KMs range. MAN can be wired or wireless. TV cable network within a city and broadband network (wired as well as wireless) are examples of MAN.

• **Wide Area Network (WAN):**The coverage scope of a WAN is spread over a large geographical areaand covers many cities, states and even countries. A WAN includes millions of computers connected located over a large geographical scale. The Internet is the largest Wide Area Network ever established.

There are three categories of wide area networks as follows:

Enterprise network: A network owned and managed by a single organization is named as enterprise network. It is the interconnected version of all the LANs of an organization/institution.

Global network: Networks of several organizations over a wide area collectively known as a global network.

Internet: A network of networks of broad area category. It is the largest network worldwide. It includes many LANs, WANs, MANs, and millions of computers. No single authority controls it; rather the local, national authority controls every segment of the Internet.

In addition to above, computer networks can also be classified as per the following characteristics:

• ***Topology*****:** topology is how computers are arranged physically i.e. the physical arrangement of computer systems in anetwork. Widely known computer network topologies are: bus topology, star topology, ring topology, and mesh topology.

• ***Architecture*****:** based on the architecture, networkscan be classified as:

- peer-to-peer network or
- client - server architecture.

Transmission Technology: according to the transmission technologynetworks can be classified as:

- Broadcast networks.
- Point-to-point or Switched networks.

In a broadcast network a single communication channel is used which is shared by all the hosts connected to the network. In a broadcast network a message sent by any member reaches to all the members in the network. This mode of transmission is known as broadcasting. The sender sends message to all the computers connected to the network, the receivers capable of and are interested to receive the message may get it. To send a packet to all the nodes (not known) in a netwok, a special address: broadcast address is used as the destination address in the packet and it is called 'Broadcasting'. To send a packet to a group of nodes (all group members are known), 'Multicasting' is used. One possible way to achieve Multicasting is to reserve one bit to indicate multicasting and the remaining (n-1) address bits contain group number. Each machine can subscribe to any or all of the groups. Some of the common broadcast networks are the packet radio networks and satellite networks etc.

1.2.2.2 Standard Internet Protocols

Some rules are necessary to make any communication successful, this set of rules is known as the protol. A protocol allows computers to talk with each other. A protocol defines the data type may be exchanged, set of commands used to exchange (send/ receive) data, and the way data to be transfered are how it is confirmed that data is received.

In human communication with spoken language also, some protocols are to be followed to communicate or share or exchange any information. Every spoken language defines set of rules to be followed while communicating. Like to make communication effective the first and basic rule to be followed in human communication is to use the language understood by both sender and the

receiver.A person from India can communicate with a person from any other contry if both of them know (speaking and listening) a common language (i.e. Hindi or English or any other). In similar way, a hardware device can connect with other hardware supporting the same protocols irrespective of their manufacturer or type of device. Like an Email sent from aniPhone can be received on an Android device by the use of acommon mail protocol used by both.
In the Internet various protocols are designed for networking applications. Like Ethernet protocol is used in wired network, IEEE 802.11 protocol is defined for wireless communication. An Internet protocol suite is a set of standard protocols used for transmitting data over the Internet.

Standard Internet protocols may be classified into four major catagories:

1. **Application layer –** Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP).
2. **Transport layer –** Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
3. **Network layer –** Internet Protocol (IP) version 4 and version 6
4. **Data link layer –** Ethernet, IEEE 802.11 (Wi-Fi)

Application layer protocols are designed to facilitate Internet users to access various Internet resources like HTTP for accessing webpages, SMTPfor sending emails etc. A number of application layer protocols are available to use. Transport layer is defines, the way data is transmitted, received and confirmation of receipt correctly. Transport layer provides two ways to transmit data: reliable data delivery and unreliable data delivery. TCP provides reliable data delivery meaning that it is confirmed that sent data is received correctly and intact whereas UDP provides unreliable data delivery meaning that data is sent and confirmation of receipt is not bothered. Network layer protocols are responsible to route the data in the Internet.

IP is responsible to provide the address of host and delivery of it to the destination host is the responbility of the routing protocols. Protocols of data link layer are responsible for node to node data delivery. Data link layer protocols are dependent on the kind of topology used, like for wired network Ethernet protocol is used whereas for wireless topology IEEE 802.11 protocol is used. Apart form discussed here many other protocols also available in the Internetworking.

1.2.2.3 Public Network & Private Network (Intranet)

A public network is an open network i.e.anyone is allowed to connect to it. Internet is the best example of public network. Though both the public and networks are there is no technical difference between a private and public network in terms of hardware and infrastructure, except for the security, addressing and authentication systems in place. there is no In a public network no or very few restrictions or access control rules are applicable to get connected and access resources of the network. In public networks least security measures are applicable therefore, it is the least secure network and one has to be very careful with respect to the security aspects while connecting to a public.

A private network is managed and controlled by an authority and access to a private network is restricted. Network of school, university or any organization are examples of private networks. In a private network the network administrator can apply rules and regulation on, how to access, what to access and how much to access the Internet. Private network is sometimes also known as Intranet. Intranet means within the network i.e.a computer network for sharing information, and computing services within an organization, with no access to outsiders.

In public networks public domain IP addresses are used whilein a private network private domain IP addresess are to be assigned to the systems. In a private network there is a requirement of NAT (network address translator) which converts private IP address to public IP address before going into the public network and vice versa. Private domain IP addresses used in a private network or Intranet can be reused in other private network but, In a public network a public domain IP address can be used on a single system. While connecting to the public network i.e. Internet, one need to obtain public IP address could be obtained from your ISP.

1.2.2.4 Accessing the Internet

Access to the Internet can be availaed using either of the three methods namely: telephone network, cable network, and wireless network.

1.2.2.4.1 Telephone Network

The PSTN (public switched telephone network) is the telephone network used for voice-communication and spread throughout the world. For data installing a separate infrastructure in parallel to the existing PSTN is not easy and a worthy solution, instead the existing infrastructure for voice communication is used for the data communication also. Hence the telephone network system is inseparable/integral part of the computer data transmission. The very first data communication over telephone lines is implemented through dial-up setup. Further its enhanced version: dedicated lines came into existence.

Figure 3: PSTN Network

Dial-Up Lines

Setting up data connection over existing analog telephone lines require to install a modem on both the sides (sender side and receiver side). Establishing a data connection over the telephone line using **dial-up setup** is as similar as establishing a voice call over the same. Before sending the computer data, a connection is required to be established which is temporary in nature between sender and the receiver. To establish the connection modem on sender side dials the telephone number of the receiver side's modem. Once the connection is established data transmission can takes place. After the completion of the data transmission, either side of the communication can turn down the connection. During thecommunication no other connections can be established between end devices and even at a time either a data or a voice communication can be possible. The speed of the data transmission is very slow over the dial-up connection. The maximum speed could be availed is 56kbps over a dial-up connection.

Some of the advantages of using dial-up connection are: Low cost, and Availability

Some of the disadvantages of using the dial-up connection are: Low data transmission speed, Requires phone line, hence could not use it for both data and telephone voice communication, connection has to be reestablished each time, and Route busy: no other connection could be established between communicating end devices.

Dedicated Lines

Connecction established using dedicated lines does not require to dial the telephone number of the other end. A dedicated line establishes a separate connection for the data from the voice connection. It is always-on type of connection i.e. there is no need to reestablisheda connection each time it is used. Dedicated lines provide a high quality connection for data transmission than a dial-up line. Dedicated lines are classified into two categories namely: analoglines and digital lines. In present scenario the digital lines are widely used to connect end users to Internet worldwide. The data transmission over digital dedicated lines is much faster than that over the analog lines.

Some of the digital dedicated lines are ISDN lines, DSL.

Integrated Services Digital Network (ISDN)

Data transmission over Integrated Services Digital Network (ISDN) is much faster as compared to the dial-up lines. ISDN also uses standard telephone lines for data transfer and was one of the popular technique for data communication. An ISDN modem is required to be installed at both the communicating ends. The end used should not be out of the range of 3.5 miles from ISDN modem of the telephone connection providing company. Thus, ISDN is suitable to urban areas only and is not an option for users located in rural areas far from the cities (where ISDN modems of telephone service providers are installed). Data can be transferred at the rate of 64kbps over ISDN. Unlike dial-up line, speech and data/information can be transmitted over same line.

Digital Subscriber Line **(DSL)**

Figure 4: DSL access to Internet

DSL is used for transmitting information at fast speed using the existing telephone lines. The user can use both telephone and computer data transmission at the same time over the same line. A DSL modem is required to be installed at the end of the line to enable the computer to connect to Internet.

Unlike dial-up line there is no need to dial the telephone number of the other end to establish the connect, rather this type of connection is always-on connection.

Asymmetric digital subscriber line (ADSL) is a type of DSL widely used for Internet connection. One of the reason of popularity of ADSL is the higher downstream (downloading) rate than the upstream (uploading) rate. Most of the Internet users use the Internet services to download the contents from Internet making it more popular.

1.2.2.4.2 Cable Network

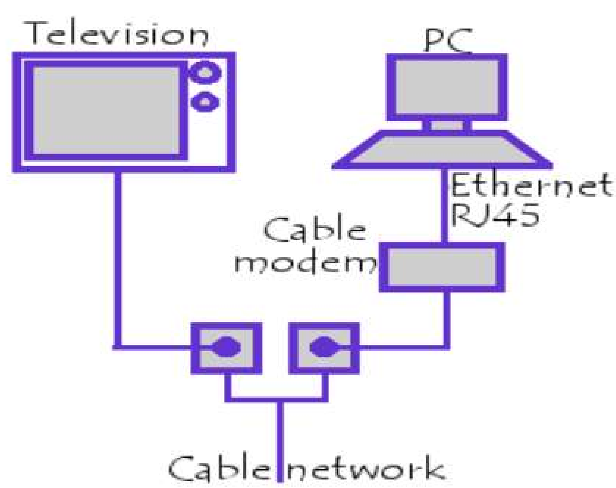

Figure 5: Cable Network

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In a cable network cables are used as transmission medium in the network. Cable Internet uses the television cable infrastructure to provide the Internet accessibility. This is also called broadband Internet access. Cable modem device is used for Internet access in the cable network. Coaxial cable is used in the cable network for connectivity. Cable Internet provides faster Internet speed of upto 1 Gbps than dial-up and DSL connections.

Cable Internet access provides last mile accessi.e. the cable network of television infrastructure is used from Internet service provider to end user. In today's scenario the cable Internet access is suppressed by fiber optical network, wireless network, and mobile networks.

Advantages of using cable Internet access: It provides high speed data transfer, it is convenient to use and easy installation, and it doesn't use telephone lines thus doesn't affect the telephone lines.

Disadvantages of using Internet access: it is costlier than the dialup and DSL connection, it is less secure as compared to the dialup or DSL connection, This service is not available with all cable Television networks.

1.2.2.4.3 Wireless Network

Wireless networks is much cheaper to install and maintain. Wireless network infrastructure does not require setup of wired network. A wireless modem/router/access point is required to establish wireless network. Many wireless technologies are available to establish wireless network. Bluetooth, InfraRed, Wi-Fi etc are some of the wireless technologies commonly used to establish a wireless network. Wi-Fi (wireless Fidelity) is one of the most widely used technology for wireless networks. A Wi-Fi enanble modem has a typical range of approx 90 mtrs. The connection speed may reach up to 1544 Kbps.

1.2.2.5 Internet Services

Using Internet services, one is able to access information available on the Intrernet. These Internet services can be classified as four major categories: Communication services, information retrieval services, web services, and eord wide web.

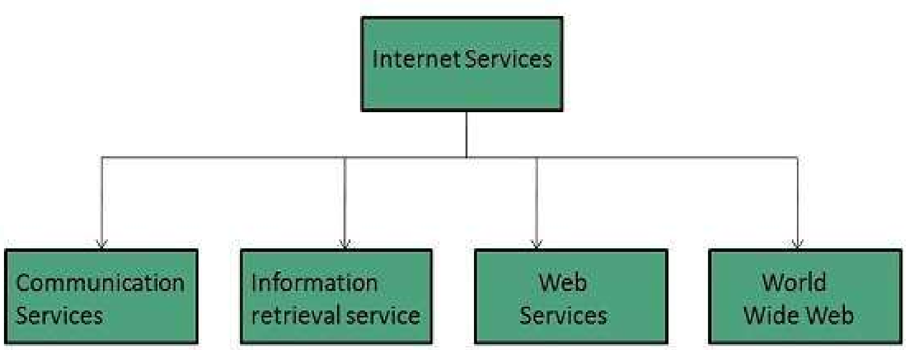

Figure 6 : Internwet Services

Communication Services

Communication services are used to exchange information over the Internet. One of the commonly used communication service is the Electronic mail. Electronic mail service is used to exhangethe messages in the form of electronic messages. Other communication services are: Telnet – used to establish and exchanges messages between remote systems, Internet Relay Chat (IRC) – it is used to communicate in real time, Internet Telephony (Voice over IP also known as VoIP): the internet users are allowed to talk using Internet services by making a call, Instant Messaging (IM): Ims allows real time chat between users and group. Eg.gtalk (google talk), Yahoo messenger etc.

Information Retrieval Services

Information retrieval services are used to download or extract information from the Internet. One of the commonly used information retrieval services in Internet is FTP (File Transfer Protocol): it enables the Internet users to transfer files to or from the Internet.

Web Services

Web applications uses web services to interact with each other.

World Wide Web

World wide web (WWW) provides a way to retrieve documents located on servers located on different geographical locations over the internet. WWW documents may include audio, text, visuals, graphics and links to other documents (hyperlinks). Hyperlinks are used to navigate among the documents.

Check your progress 1

Q.1 What is Internet ? Explain.

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Q.2What is Internet Protocol? What internet protocol is classified?

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Q.3 Define DSL and ADSL?

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1.3 NETWORK TOPOLOGY

Topology of a network is defined as: the physical and logical arrangement of the nodes in the network. How nodes are interconnected defines the topology of the network. Topology could be defined for a wired network as devices are wirelessly connected and free to move and change their physical locations. Topology of the network directly affects the performance of the system. Before deploying a networkit is very important to select appropriate topology to be installed. Topology of the network not only affects the performance of the network but also the installation cost of the network. The parameters to be considered in selection of the topology are:the size of he network, number of nodesin a network, the purpose of network, and capacity of fault tolerance etc. In computer networking network topologies are classified into four majoe categories namely: Bus topology, Ring Topology, Star topology and Mesh topology.

Bus Topology

In a bus topology nodes /computers are connected to a common link (also nown as backbone link). Sometimes bus topology is also called a line or backbone topology. The data flows in single direction on the common channel. In a bus topology at a time only one computer can transmit the data. If more than one computer transmits data at a time, a collision occurs. One of the disadvantage of bus topology is single point of failure i.e. if the backbone link or the common link gets faulty the overall topology gets down.

Figure 7: Bus Topology

Some of the advantages of bus topology :

- Installation wise bus topology is cheaper as compared to others.
- Easy to install
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Some of the disadvantages of bus topology:

- If the backbone link is in fault, the communication of whole topology fails.
- At a time only one computer can transmit the data, if this rule is not followed collisions occur.
- Any information available on the common link is accessible to all connected computers, leading to unsecure communication.

Star Topology

All the computers/devices are connected to a central switch/hub/server forming a star like topology. All the communication in star topology take place through the central node. The centralized node can act as a simple forwarding device or can also act as an intelligent firewall / filtering device.

Figure 8: Star Topology

Some of the advantages of star topology :

- Easy to install and extend.
- Robust toward fault tolerance.
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Some of the disadvantages of star topology:

- Single point of failure, i.e. if the central node fails, overall system crash.
- Performance of the topology is depended on the central node's performance.

Ring Topology:

Each device/computer is connected with two devices forming a ring type of tructure. In general the data can move in one direction in the ring, but by installing another cable the data can move in reverse direction also. A token message keeps on rotating in the ring topology.

A node having some data to be transmitted, hold the token message and transmits the data, after completion of data transmission, the token is forwarded to next node. If a node does not have any data to send, it simply forwards the token to next node. Without holding the token data transmission is not allowed in ring topology.

Figure 9: Ring Topology

Some of the advantages of ring topology are:

- Minimum chances of data collirion, as transmission is restricted with the help of token message.
- It is cheap to install.
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Some of the disadvantages of ring topology are:

- Fauld diagnosis is difficult in ring topology.
- Extension of the topology by adding a new device or removing an existing device is not easy.

Mesh Topology:

In a mesh topology each device/node is directly connected to every other device/node. It is one of the most reliable and fault tolerant network topology. To connect n device/nodes in a mesh topology, it requires n(n-1)/2 number of cables.

Figure 10: Mesh Topology

Some of the advantages of mesh topology are:

- Mesh topology is highly robust.
- Easy to diagnose the fault.
- As data is sent to destination directly without any intermediate node, hence data transmission is secure in mesh topology.

Some of the disadvantes of Mesh topology:

- Installation of mesh topology is most costlier.
- Maintenance is not easy.
- For a large number of devices/nodes it is very difficult to manage mesh network.

1.4 NETWORK TOPOLOGY

The computer networking involves both hardware and software components. Networking system also deals with embedded chip level prograaming and electric pulses. Overall the computer networking system is very complex to understand, implement and to extend. Further to make it easy to learn, extend and to provide interoperability the network models were developed with hierarchical approach with multiple layers. Each layer is independent of other layers and responsible for task assigned to it. A layer can communicate to layers just above it and just below it hence extension in any layer affects only the layers directly in contact with it. In general There are two network models defined in the networking: OSI model and TCP/IP model.

- - OSI Models: OSI is Open System Interconnection. It is also known as reference model for networking. OSI model is not exactly based on the working of the networking instead it was designed for understanding purpose of the networking. OSI model is seven layer model. It was proposed by ISO (Internation Organization for Standards) in 1984 after the TCP/IP model.

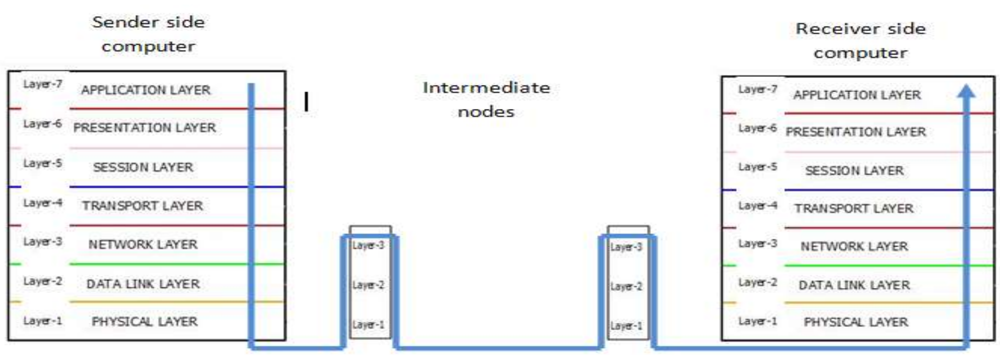

Figure 11: OSI reference model

- **Layer 1 - Physical Layer:** This layer is responsible for defining hardware related standards. Standards like cableing types, their codings, connectors, ports etc are defined at this layer. Parameters related to physical transmission of data over medium i.e. signaling, pulse rate, bit rate, baud rate, power output etc. Layer 1 devicev is Hub.

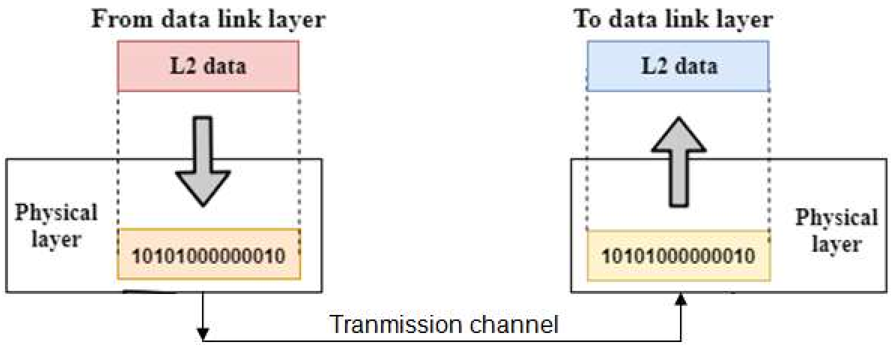

Figure 12 : Physical layer

Functions of a Physical layer:

- **Line Configuration:** how devices are physically connected to each other using wires or wireless channel is defined by line configuration.
- **Data Transmission:** transmission mode: simplex, or duplex (half or full) between two endsis defined by physical layer.
- **Topology:** physical arrangement of devices/ nodes defines the topology which is part of physical layer.
- **Signals:**Data is transmitted on physical medium in the form of electrical signals. The type of signals used to transmit the data is determined at physical layer.

- **Layer 2 –** Data Link Layer (DLL): This layer deals with both the hardware part and software part. This layer is mainly implemented in the Network Interface Card part of the computer. Data transmission between node to node is responsibility of this layer. It is also error control and flow control of traffic between two ends of the link. Frames are created specific to the underlying link. Physical address also known as the Media Access Control (MAC) address is used for recognizing nodes in the network for data delivery. MAC address is of 48 bits and is the property of the NIC. Out of 48 bits first 24 bits are recognizing the vendor created the NIC and rest 24 bits are unique for NICs of the vendor. Protocols used on this layer are: Ethernet, ATM etc. Layer 2 devices: Switch, Bridge etc.

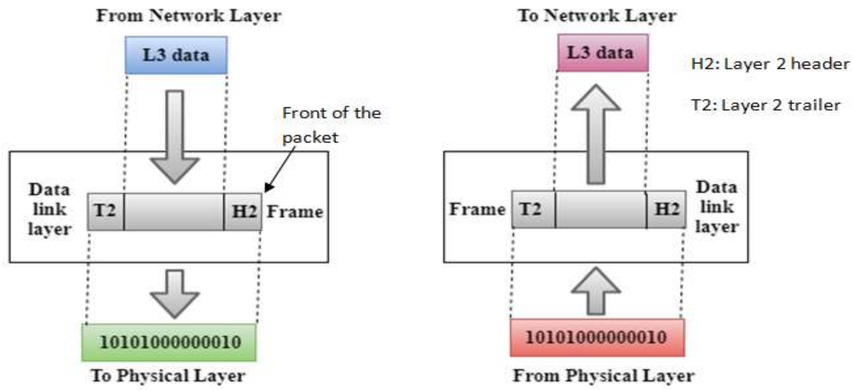

Figure 13: Data Link Layer

Functions of the Data-link layer

- **Framing:** DLL creates the physical medium specific frames by adding header and trailer to it. DLL is also responsible for creating frames from the physical bit stream received from physical layer on destination side.
- **Physical Addressing:** DLL includes physical address or MAC address of sender and receiver into header of the frame.
- **Flow Control:** DLL is responsible for flow control across the link.
- **Error Control:** DLL is also responsible for error control by adding a CRC (Cyclic Redundancy Check) value into the trailer part of the frame. Frames reported in error are retransmitted.
- **Access Control:** DLL allows transmission over a shared channel without collisions.

Layer 3 – Network Layer: Network layer is responsible for assigning IP addresses to networked devices. It is also responsible for best path selection from source to the destination for data deliveryi.e.routing in the network.It provides connectionless services to the upper layers. Protocols used at this layer are: Internet Protocol (IP), IPv4 and IPv6, ICMP. Network layer devices are: Routers.

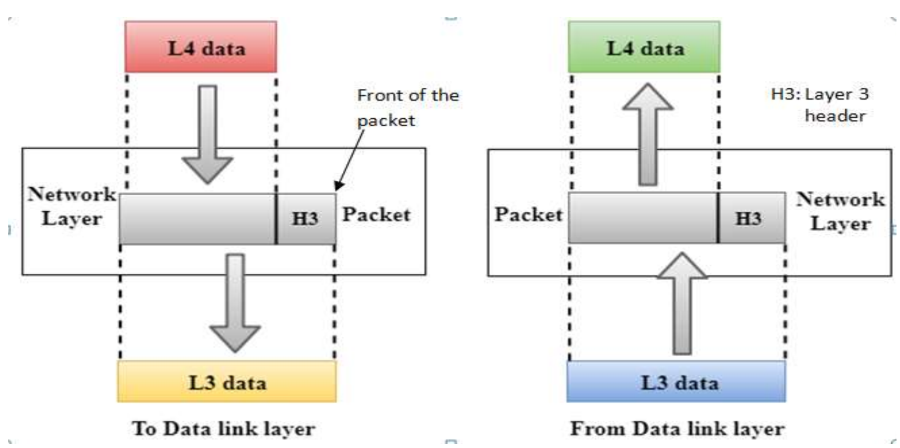

Figure 14: Network Layer

Functions of Network Layer:

- **Internetworking:** Network layer is responsible for data delivery between end hosts with the help of logical addressing i.e. IP addressing.
- **Addressing:** Network layer re responsible for assignment of the IP addresses to the networked devices. IP address is used to recognize a device in the Network.
- **Routing:** Routing is one of the major and important taskof this layer. With the help of routingit determines the best optimal path between source and the destination.
- **Packetizing:** It receives the data from upper layer i.e. transport layer and adds its control information as header to form packets.

Layer 4 – Transport Layer: Transport layer is responsible for identification of the applications/ services by assigning the port numbers. End-to-end data delivery is the responsibility of the transport layer using sockets. A socket is combination of two entities: port number and IP address. This layer provides two kinds of data delivery services: connection oriented (reliable data delivery) and connectionless (unreliable data delivery). It is the 1st layer where user data is breaked down into smaller segments. This layer receives formatted user data from upper layer i.e. session layer and adds its control information in the form of header to create segments. Protocols used at this layer are: **Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).**

TCP provides connection oriented reliable data delivery. It establishes a logical/virtual connection between end hosts before transmitting the data. TCP is also able to deliver segments in order (same as sent) and ensures the delivery of all the segments without error. Unlike TCP the **UDP**provides unreliable connectionless services. It does not guarantee the data delivery. It is the responbility of the application to check the correctness of the data received also it is not responsible for ordered delivery of the segments. This protocol is used for the application which can tolerate the data loss. Data delivery using UDP is faster than TCP with less control information overhead. The header inserted by UDP is smaller than that of TCP.

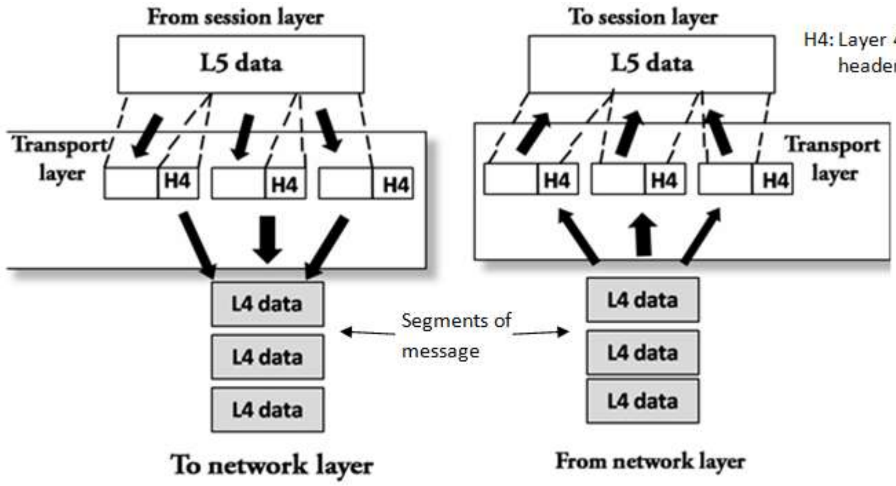

Figure 15: Transport Layer

Functions of Transport Layer:

- **Service-point addressing:** transport layer assignes a unique number: port number, to each computer networking application (even this port number is assigned to each of the tab of the web browser). This port number is added into header as control information. As discussed earlier the network layer is responsible to deliver the data to the computer (with the help of IP address) whereas the transport layer is responsible to collect the data from network layer and handover it to the correct application/process within the computer (at destination.
- **Segmentation and reassembly:** On the sender side transport layer divides the formatted user message into smaller segments, and assigns a sequence number that uniquely identifies each segment of the message. This sequence number is used to reassemble these segments on the detination side. Also this sequence number is used to confirm whether all the sengments are delivered or not and are inorder or out of order.
- **Connection oriented services:** Transport layer provides connection oriented services with the help of TCP. A logial end-to-end (application to application or socket to socket) connection is established before sending the user data. These services are used to provide reliable inorder data delivery services.
- **Connectionless services: using UDP** transport layer can deliver data without bothering about the correctness and order of the segments received. It is the responsibility of the application layer to rearrange the segments to bring segments in order.
- **Flow control:** This layer provides flow control mechanism between end-to-end hosts. TCP windows are used to provide flow control facility.
- **Error control:** Unlike DLL which provides error control across the link, transport layer provides the error control facility between ent-to-end hosts.

- **Layer 3 – Session Layer:** Session layer is responsible for maintaining sessions between hosts. It establishes, maintains and synchronizes the communication between end hosts. Ex. once user credentials are verified, the session layer at remote host keeps this session alive according to some predefined conditions and doesn't prompt for user credentials again until this session expires.

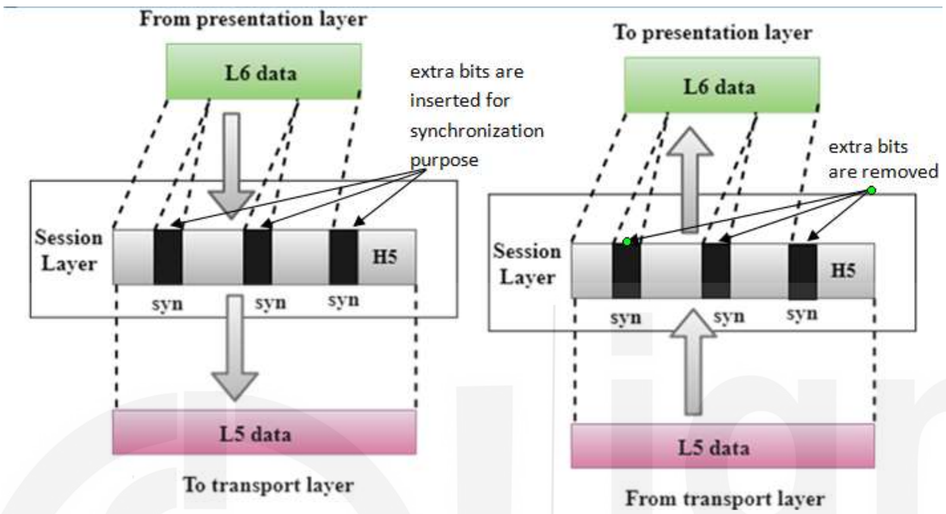

Figure 16: Session Layer

Functions of Session layer:

- **Dialog control:**Session layer provides two types of communication: half duplex and full duplex. It also records whose turn is it to transmit the data.
- **Synchronization:** Session layer is also responsible for synchronizing the data transmission by adding checkpoints while transmitting. If some occurs in between transmission, the system is rolled back to the checkpoint and retransmission begins from here again.

- **Layer 2 – Presentation Layer: On the sender side** this layer formats the message received from upper layer i.e. application layer and providing this formatted message to the underlying session layer. It defines the syntax and semantics of the message exchanged. On the destination side this layer receives the data from underlying layer i.e. session layer and reverses the operations of formatting to get the user raw message and provides it to the concerned application. Sometimes this layer is also called as the syntax layer.

Functions of Presentation layer:

- **Translation:** Session layer translates data from sender system's format to a generic format and reverse translating from the generic format to destination system's format.
- **Encryption:** Encryption and decryption of the message is the responsibility of presentation layer. Message encryption is performed on the sender side for security purpose and the message is decrypted on the destination side at presentation layer before handling it to the concerned application.
- **Compression:** At session layer data compression process takes place on the sender side and on the receiver side message will be decompressed based on the compression techniques used (lossy or lossless). Images are compressed with jpeg technique in the Internet.

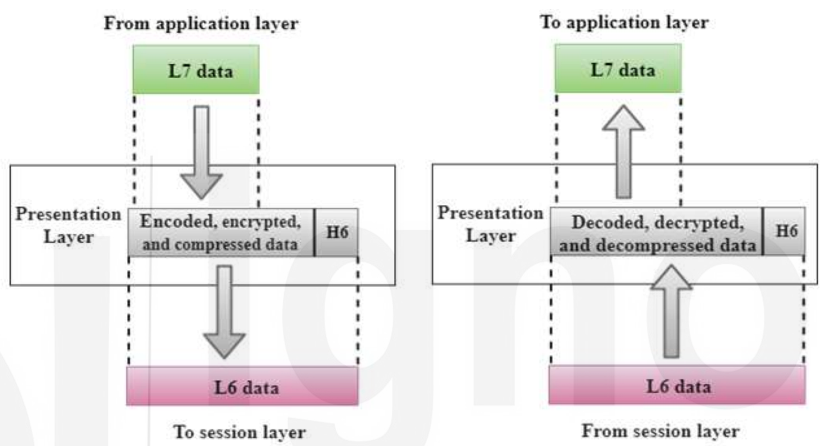

Figure 17: Presentation Layer

- Layer 1 – Application Layer: Application layer provides an interface to the user to interact and avail the Internet services. To be precise the application layer is not an application, rather it carry out the functionalities of this layer. Some of the application layer protocols are: HTTP, FTP, SMTP,Telnet etc.

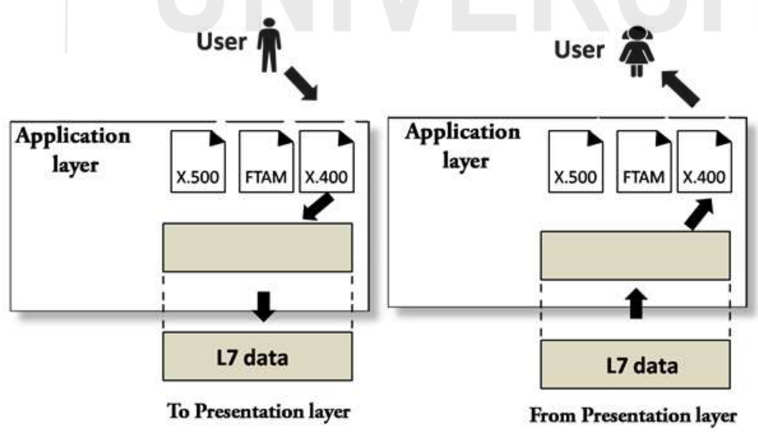

Figure 18: Application Layer

Functions of Application layer:

This layer provides performesuser oriented services. Some of the commonly used application layer services are:

- **File transfer, access, and management (FTAM):** Application layer provides functionalities to download, upload and manage files places on a remote system through Internet.
- **Mail services:** Application layer is responsible to provides the facility of sending, receiving and storing emails in the Internet.

- TCP/IP Model

The TCP/IP networking model was introduced before the OSI reference model. The TCP/IP model was abstract in nature as compared to the OSI model.The TCP/IP model has four layers namely: application layer, transport layer, Internet layer and link layer. Layers of TCP/IP model are recognized by their names unlike as recongnized by their layer number in OSI reference model. TCP/IP model is more closely defined to the actual working of the networking. Layers 7, 6 and 5 of OSI reference model are combined and nemed application layer in TCP/IP model. Rest of the layers are similar in both the models.

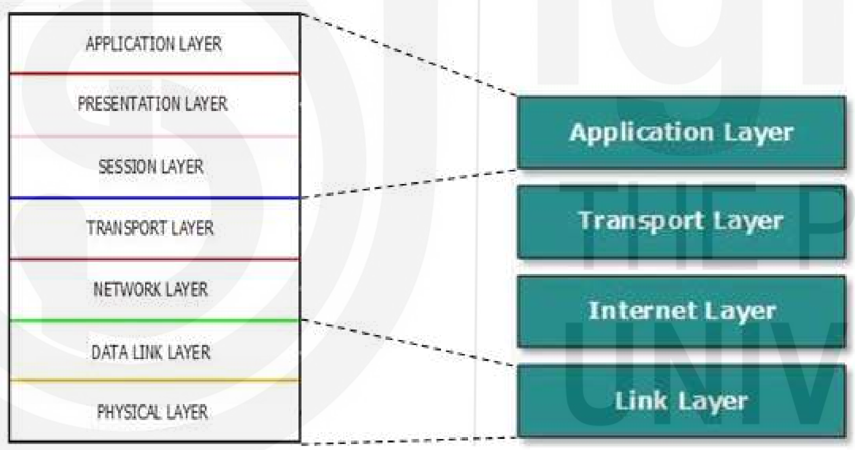

Figure 19: TCP/IP Networking Model

- **Application Layer**: It defines the interface for the user to access the Internet services. This layer is also responsible for formatting of the message to be transmitted and maintenance of the sessions. Ex. FTP, SMTP, HTTP etc.
- **Transport Layer**: Transport layer is responsible for process to process data delivery. s Transmission Control Protocol (TCP) and User datagram Protocol (UDP) are two major protocols of the transport layer.
- **Internet Layer**: Internet layer is responsible for host addressing and recognition in the Internet by assigning IP address. The routing is defined at this layer.

- **Link Layer**: Link layer is combination of DLL and Physical layer of OSI reference model.

Check your progress 2

Q.1 Explain World Wide Web (WWW) ?

- - - - -- -

- -

Q.2 What is ring and star topology? What are their advantages?

- - - - -- -

- -

Q.3What are different functions of Data link Layer?

- - - - -- -

- -

Q.4Explain TCP/IP model with layer functionality.

- - - - -- -

- -

Q.5 Access network connects to the Internet via tier -3 ISPs also known as?
a) Connection ISPs
b) Access ISPs
c) Internetwork ISPs
d) ISDNs

Q.6 Protocol used in wireless topologies?
a) IEEE 802.11
b)IEEE 802.12
c)IEEE 802.13
d) None

Q.7 A Wi-Fi enabled modem has a typical range of approx?
a) 490 meters.
b)290 meters.
c) 90 meters.
d) 250 meters.

Q.8 Which of the following topology is also called a line or backbone topology?
a) Ring
b) Star
c) Bus

d) Mesh

Q.9 Which layer is responsible for assigning IP addresses to networked devices?

a) Transport Layer

b) Data Link Layer

c) Physical Layer

d)Network Layer

1.5 SUMMARY

In this unit we have discussed about the Internet. Internet is network of networks. It is the worlds largest network. Further, we have discussed about the public and private networks. Public networks are open to all i.e. anyone can access a public network. A private network is restricted in nature i.e. authentication parameters are required to get access of a private network. Internet accessed methods are also deiscussed in this unit. The Internet could be accessed through telephone lines, cable network and wireless network. Using telephone lines Internet can be accessed using dial-up setup, ISDN or DSL/ADSL setups. Network Topology is defined as the physical or logical arrangement of the devices in a network. Commonly four types of network topologies namely: Star topology, Bus topology, Ring topology and Mesh topology are discussesin this unit. The two network models namely: OSI model and TCP/IP model are discussed in this unit.

1.6 SOLUTIONS/ANSWERS

Answers to CYP 1:

Ans 1: Internet is a network of networks of broad area category. It is the largest network worldwide. It includes many LANs, WANs, MANs, and millions of computers. No single authority controls it; rather the local, national authority controls every segment of the Internet.

Ans 2: An Internet protocol suite is a set of standard protocols used for transmitting data over the Internet. Standard Internet protocols may be classified into four major catagories:

1. **Application layer –** Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP).
2. **Transport layer –** Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
3. **Network layer –** Internet Protocol (IP) version 4 and version 6
4. **Data link layer –** Ethernet, IEEE 802.11 (Wi-Fi)

Ans 3: DSL is used for transmitting information at fast speed using the existing telephone lines. The user can use both telephone and computer data transmission at the same time over the same line. A DSL modem is required to be installed at the end of the line to enable the computer to connect to Internet. Unlike dial-up line there is no need to dial the telephone number of the other end to establish the connection, rather this type of connection is always-on connection.

Asymmetric digital subscriber line (ADSL) is a type of DSL widely used for Internet connection. One of the reason of popularity of ADSL is the higher downstream (downloading) rate than the upstream (uploading) rate. Most of the Internet users use the Internet services to download the contents from Internet making it more popular.

Answers to CYP 2:

Ans 1: World Wide Web (WWW) provides a way to retrieve documents located on servers located on different geographical locations over the internet. WWW documents may include audio, text, visuals, graphics and links to other documents (hyperlinks). Hyperlinks are used to navigate among the documents.

Ans 2: In a ring topology, each device/computer is connected with two devices forming a ring type of structure. In general the data can move in one direction in the ring, but by installing another cable the data can move in reverse direction also. A token message keeps on rotating in the ring topology. A node having some data to be transmitted, hold the token message and transmits the data, after completion of data transmission, the token is forwarded to next node. If a node does not have any data to send, it simply forwards the token to next node. Without holding the token data transmission is not allowed in ring topology.

In a star topology, all the computers/devices are connected to a central switch/hub/server forming a star like topology. All the communication in star topology take place through the central node. The centralized node can act as a simple forwarding device or can also act as an intelligent firewall / filtering device.

Advantages of ring topology:

- Minimum chances of data collision, as transmission is restricted with the help of token message.
- It is cheap to install and configure.

Advantages of star topology:

- Easy to install and extend.
- Robust toward fault tolerance.

Ans 3: Data Link Layer perform following functions:

- **Framing:** DLL creates the physical medium specific frames by adding header and trailer to it. DLL is also responsible for creating frames from the physical bit stream received from physical layer on destination side.
- **Physical Addressing:** DLL includes physical address or MAC address of sender and receiver into header of the frame.
- **Flow Control:** DLL is responsible for flow control across the link.
- **Error Control:** DLL is also responsible for error control by adding a CRC (Cyclic Redundancy Check) value into the trailer part of the frame. Frames reported in error are retransmitted.
- **Access Control:** DLL allows transmission over a shared channel without collisions.

Ans 4: The TCP/IP networking model was introduced before the OSI reference model. The TCP/IP model was abstract in nature as compared to the OSI model.The TCP/IP model has four layers namely: application layer, transport layer, Internet layer and link layer. Layers of TCP/IP model are recognized by their names unlike as recongnized by their layer number in OSI reference model. TCP/IP model is more closely defined to the actual working of the networking. Layers 7, 6 and 5 of OSI reference model are combined and named application layer in TCP/IP model. Rest of the layers are similar in both the models.

Layer Functionality:

- **Application Layer**: It defines the interface for the user to access the Internet services. This layer is also responsible for formatting of the message to be transmitted and maintenance of the sessions. Ex. FTP, SMTP, HTTP etc.
- **Transport Layer**: Transport layer is responsible for process to process data delivery. s Transmission Control Protocol (TCP) and User datagram Protocol (UDP) are two major protocols of the transport layer.
- **Internet Layer**: Internet layer is responsible for host addressing and recognition in the Internet by assigning IP address. The routing is defined at this layer.
- **Link Layer**: Link layer is combination of DLL and Physical layer of OSI reference model.

5. B, 6. A, 7. C, 8. C, 9. D

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UNIT 2 DATA TRASNMISSION BASICS & TRANSMISSION MEDIA

2.0 INTRODUCTION

Data communication not only involves transfer of information but is successful only when the information is interpreted correctly by receiver.
For any communication to be successful some set of rules has to be followed. Set of rules is called protocol. As the communicating devices most of times are heterogeneous in nature i.e. may be different in hardware or software or operating system plateformetc that is why certain rules has to be followed to bring them all on a common platform to make them capacble of communicate and understand the information. As we know that to transmit the data from one system to another separated apart from each other there is a need of medium/channel. Data transmission can takes place either through wired media or wireless media. A communication protocol is designed considering following parameters:

- Media/channel used in transmission;
- Rate at wich data to be transmitted (generally in bits per second);
- Mode of data transmission;
- Attenuation, Delay, Noise etc faced during transmission

In data communication these are used every where, whether it is comparing efficiency of protocols for data transmission, routing, security means etc. To understand these in better way we have to understand commonly used terminologies and basic concepts of data transmission first, and that is what we are going to learn in this unit.

In section 2.1 data communication terminologies like Channel, Baud, Bandwidth, and Frequency. In section 2.2 various data transmission modes are discussed. Section 2.3 covers analog and digital data transmission.Transmission Impairments: Attenuation, Delay Distortion, Noise, Signal to Noise ratio, and Concept of Delays are discussed in section 2.4. Guided and unguided transmission media are covered in section 2.5. Section 2.6 coveres wireless transmission methods: Microwave Transmission, Radio Transmission, Infrared and Millimeter Waves. Section 2.7 discusses the Wireless LAN technology. In section 2.8, summary is covered.. Some of the important Solutions/Answer covering the entire unit are discussed in section 2.9.

2.1 OBJECTIVES

After completing this unit, one should be able to:

- understand the Data Communication Terminologies;
- understand Data Transmission Modes;
- understand Analog and Digital Data Transmission;
- understand the terms Attenuation, Delay Distortion, Noise, Signal to Noise ratio etc;
- understand the Concept of Delays;
- differentiate Guided and Unguided Transmission Media;
- understand the Wireless Data Transmission Mode.

2.2 DATA COMMUNICATION TERMINOLOGY

As discussed in section 2.1 Data communication is not just to transfer the data between sender and receiver but also that the interpretation of the data should be same and correct by both the sednder and the receiver. Further we will be looking after the terminologies commonly used in data communication here:

2.2.1 Channel

A channel is the medium used to transmit the information between sender and receiver. In other words channel is the path through which the exchange of information between devices takes place. Channel is classified into two categories: wired and wirless. In metal wired channel the electrons present within the metal carry the information. In fiber optics based guided media concept of total internal reflection of light is used to transmit the information in a fiber glass cable. Wired channels are guided media i.e. as the electrons can move in the metal in restricted way in certain direction carrying the information, whereasin the wireless channels the air particals are responsible for data transmission.

Every channel has the channel capacity which limits its maximum data carrying capacity over a certain time period. Based on the type of information could be propagated through channel, there are two types of channels used in data communication namely: Analog and Digital. Analog form of data is transmitted through analog channels and digital data is transmitted through digital channel.

2.2.2 Baud

It is a unit of data transmission speed. Symbol rate or modulation rate of a signal is measured in bauds. Symbole rate is one of the elementsconsidered to measure the data transmission speed of a data channel.

In a second how many times a signal changes its state is known as the baud. For digital signals with two states, baud and bita per second are equal i.e. 1 Bd = 1 bit/s.

The symbol 'Bd' is used to represent the unit baud. For example: Transmission at baud rate 100 *Bd* means, 100 symbols are are transmitted in a second i.e. a symbol is transmitted for 100^{th} fraction of a second.

Baud rate is related to bit rate as following:

Bit rate = Baud rate x bits in a Baud

2.2.3 Bandwidth

Bandwidth is the data carrying capacity of the channel/media per unit time (It can be defined for the bus within a computer or the channel of computer network). Bandwidth of a channel is the amount of data or signals transmitted in a time interval. Bandwidth of a channel depends on the length of the channel, media considered, and the signaling method used. More is the bandwidth higher is the throughput of the medium. Bandwidth ofmedium is the range of the frequencies used to transmit the data i.e. the difference between highest and the lowest frequencies the medium can carry. The bandwidth for digital devices is typicallyconsidered as number of bits per second whereas, expressed in Hertz (Hz) or number of cycles per second for analog devices.

2.2.4 Frequency

Frequency is the rate of repeatition of an event per second of duration. Unit of frequency is Hz which is number of cycles per second.
In data communication, the frequency of a signal is number of cycles completed by the signal per unit time.Some times there is confusion between period and frequency. Period and frequency are exactly opposite of each other. Period is defined as the minimum time in which a signal repeates itself and denoted as seconds per cycle whereas, frequency is the number of periods a signal completes within a specified time duration and denoted as cycles per second.

$$f = 1/T$$

For example, consider the following figure:

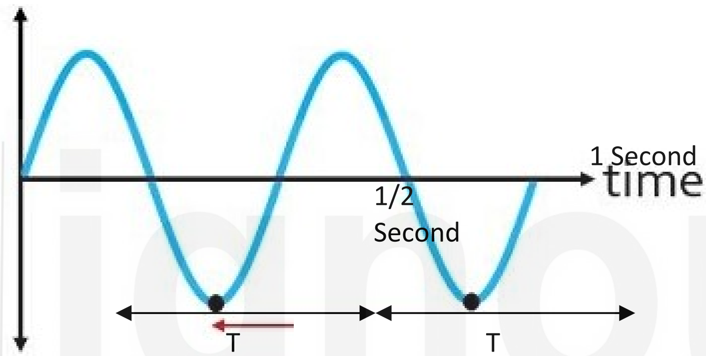

Figure: Time Period and Frequency of a Signal.

In this figure the signal repeates itself after ½ second so, the time period (T) of the signal is ½ second.
Frequency of this signal is: the number of cycles completed in a second i.e. the signal completes 2 cycles in a second hence, the frequency (f) of the signal is 2 Hz.

2.3 MODES OF DATA TRANSMISSION

Data transmission mode is how data is transferred from sender to receiver over a communication channel in the network. Mode of transmission defines the direction of flow of data. Based on the mode of data transmission a communication can be classified as:

- Simplex Mode,
- Half duplex mode and
- Full duplex mode.

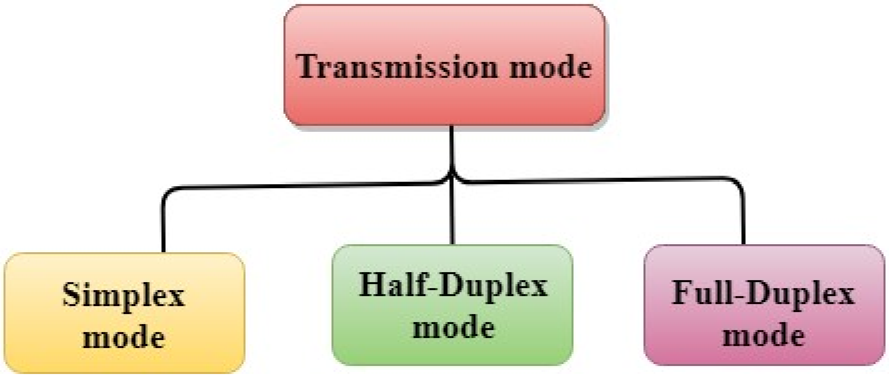

Figure: Data Transmission Modes

Other ways of data transmission are: Serial transmission and Parallel transmission. Further serial transmission is classified into three categories namely: Asynchronous, Synchronous and Isochronous transmission.

2.3.1 Simplex mode, Half Duplex mode and Full Duplex mode

Simplex mode of data transmission:

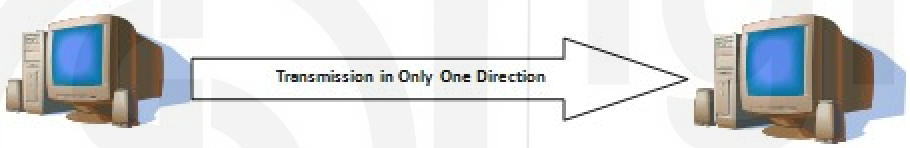

Figure:Simplex Mode

In Simplex mode of data transmission, the data through the channel can be transferred in one direction only. This mode of transmission is useful when one device always acts as the transmitter and another as the receiver. The channel used in radio (AM/FM) communication is type of simplex channel where the radio station always transmits the signals and the listeners always receives the signal. In computer system the keyboard, mouse, printer, scanner and monitor also uses the simplex mode transmission, data is always transferred in one direction only. Simplex mode of data transmission is not very useful in computer networking as data is to be sent and received in both ways. One of the advantage of simplex mode is that the station (acting as transmitter) can utilize the full bandwidth of the channel. Simplex mode is not suitable for intercommunication among systems due to the constraint of unidirectional transmission.

Duplex Mode of data transmission:contrast to the simplex mode in duplex mode of transmission data is allowed to flow in both the direction over the channel. Further duplex mode of transmission is classified into two categories namely: Half duplex mode and Full duplex mode.

Half Duplex Mode of Data Transmission: In half duplex mode of transmission the data can flow in both the direction but can be allowed in one direction at a time i.e. at a time one of the communicating devices acts as transmitter and another as receiver. Both cannot act as transmitter simultaneously.
Unlike the simplex mode in which one way communication takes place i.e. one device acts as a transmitter and another as receiver, the half-duplex is two-way communication, but only one device is allowed to send data at a time.
The full bandwidth is used by either of the communicating devices. One of the example of half duplex communication is the walkie-talkie, in which one end speaks and another listens, both are not allowed to transmit voice signals simultaneously.

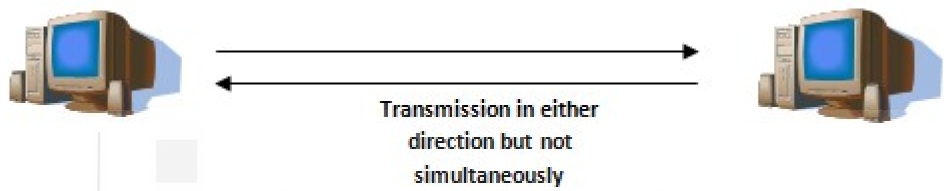

Figure: Half Duplex Mode

Full Duplex Mode of Data Transmission: In full duplex mode both the devices can act as transmitter and receiver simultaneously. Both the ends should be capable of transmitting and receiving the data simultaneously. In this mode of transmission the data can flow in both directions at a time. The bandwidth of the channel is divided between both end devices. The most popular full duplex communication is the telephone network where both ends can transmit and receive voice signals simultaneously.

Figure: Full Deplex Mode

2.3.2 Serial and Parallel Communication

Digital devices works on digital data in the form of binary bits. Data is transmitted among digital devices in the form of binary bits in two ways namely: serial transmission and parallel transmission. In serial transmission bits are transmitted in serial manner i.e. one after another through a channel whereas, in parallel transmission bits are transmitted in parallel manner i.e. multiple bits are transmitted simultaneously through multiple channels.

Serial data communication:

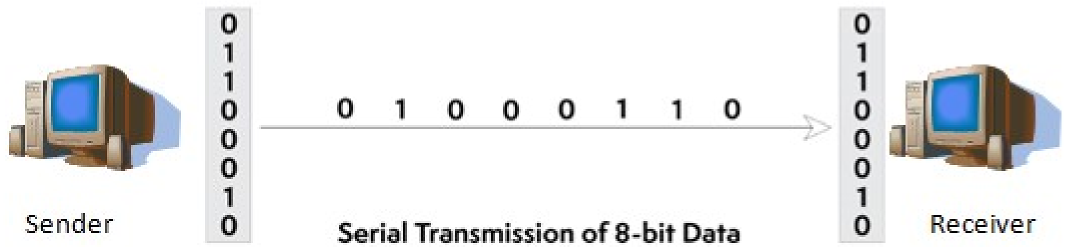

Figure: Serial Data Transmission

As shown in figure, in serial transmission between sender and receiver the data is sent bit by bit on the channel. In order to interpret the data correctly both ends mutually agree on transmitting the data from LSB (least significant bit) to MSB (most significant bit). On the sender side an extra hardware component is required to convert the bytes of characters (i.e.parallel) to serial. Simillarly on the receiver side some mechanism is to be applied to convert the received serial data into parallel to form the character bytes as sent. Serial transmission is most suitable for transferring small amount of data over long distance. Serial transmission further classified into two categories: asynchronous transmission, synchronoustransmissison and isochronous transmission which are discussed in section 2.3.3.

Parallel data communication: In parallel communication multiple bits are transmitted in parallel over multiple channels simultaneously. Data transmission through parallel data communication is much faster in comparision to transmitting the data through serial communication methods.

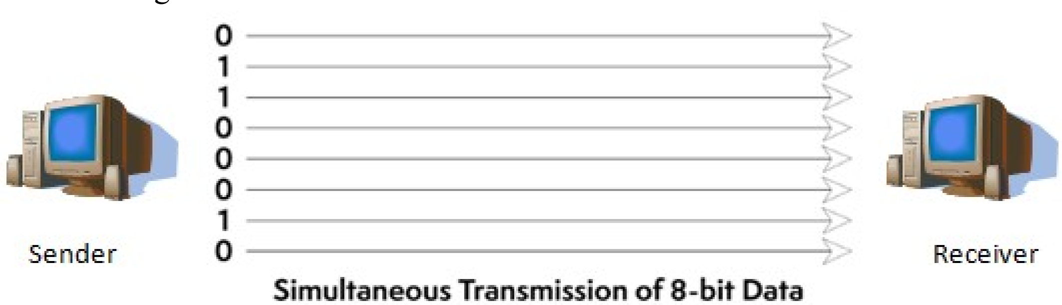

Figure: Parallel Data Transmission

In parallel data communication many bits are transmitted simultaneously hence, it is very important to note that the speed of transmission of bits on parallel channel should be always in synch with the speed of receiving of parallel bits on the receiver side. Due to this constraint parallel communication is usually not suitable for long distances instead is suitable for transmission over short distances. The data transmission of takes place in parallel making it faster, it is best suitable for transferring huge amount of data over short distances.

2.3.3 Synchronous, Asynchronous and Isochronous Communication

In general serial data transmission technioque is commonly used to transfer data in Internet over long distance. The biggest issue of serial communication is the synchronization of the transmission speed of sender and the receiving speed of the receiver. If the transmission speed and receiving speed of sender and receiver mismatches, the receiver will not be able to interpret the received bit stream correctly. In serial communication another issue is that, how to correctly detect the end of a character and beginning of next character. To synchronize sender and receiver some mechanism is to be applied. In general sender and receiver are synchronized with three methods namely: Synchronous, Asynchronous and Isochronous Communication.

Synchronous Communication: An external agent (a clock) regulates the data transmission. In synchronous way of data communication, data is transmitted in the form of frames or chunks instead of characters. A special signal named: synch signal is transmitted before sending the actual data notifying the receiver about transmission of a new frame to get synchronized with sender. Sync signal, in general is a unique bit pattern such that it cannot be a part of user data to avoid confusion between synch signal and user data. Similar to the start of the frame signal an end of the frame signal is also used to indicate that where a frame ends. Seral synchronous transmission is suitable for transferring large amount of data at a high speed. For transferring small size frames due to the overhead of start and end of frame signals the utilization of channel is not optimum.

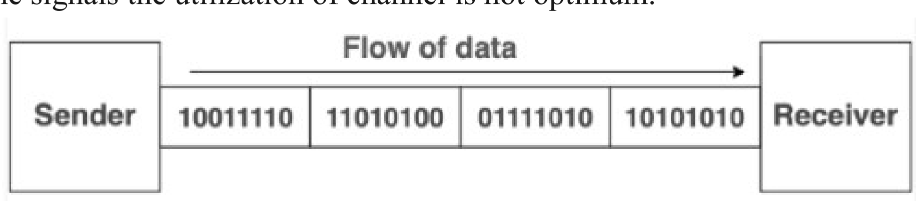

Figure: Synchrnonous Data Transmission

Synchronous transmission is preferred where characters are generated at irregular intervals because the characters are not simply transmitted as they received, they will be holded back to collect enough data to form the frame. A complex circuitry is to be installed in synchronous transmission making it expensive and difficult to implement.

As the data it transmitted in the form of frames or blocks, if any bit is found in, the whole frame or block is to be retransmitted. Cyclic Redundancy Check (CRC) error control technique is used in synchronous transmission instead of parity bit due to its capability to transmit long size frames.

Asynchronous Communication: No external agent regulates the data transmission. In synchronous communication data is sent character by character. Start and stop bits are added to each character forming a frame before sending. On receipt of start bit the receiver adjusts itself in accordance to the timing of the transmitted signal. A sender can transmit the frame at any time. In asynchronous transmission, the size of frames is should be kept as small as possible. Asynchronous communication is commonly used to transmitcharacters generated at irregularintervals. When user inputs characters with keyboard to the computer is a typical example of asynchronous communication. As the size of frame is small so, the parity bit is used to handle error control. A frame is structured with following components:

1) **Start bit:** it enables the receiver to synchronise itself with the frame (message) sent.
2) **Data Bits:** these are the actual user data to be transmitted.
3) **Parity Bits:** it is optional to use and is used to handle error encountered during transmission. Parity bit is capable of detecting single bit error only.

4) **Stop bit:** it enables receiver to detect the ned of a frame.

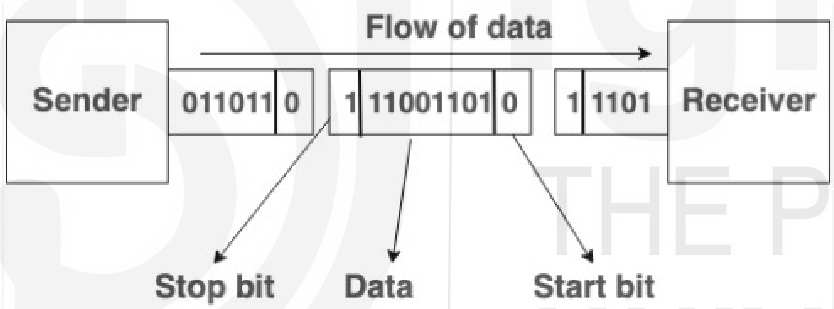

Figure: Asynchronous Data Transmission

In synchronous transmission, when the data is received by the receiver it must be handled immediately without any delay. Whereas, in asynchronous data transmission the receiver is not bound to handle the reveived data in real time instead a buffe is placed at the incoming line. Today, irrespective of serial or parallel communication, most of the transmission takes place in asynchronous manner.

In asynchronous transmission no special hardware is required to be installed makes it simple, inexpensive technique. Asynchronous communication is suitable for transmitting small size frames generated at irregular intervals (for example, characters generated by user from a keyboard).

In asynchronous communication a frame is generated from a character so, if an error occurs in a frame only a character is retransmitted contrast to the synchronous communication by restransmitting the whole block or frame (comprising many characters). The insertion of extra bits (start bit, stop bit, and parity bit), the overhead to the channel is more degrading the overall performance of transmission hence, asynchronous communication is not suitable for transmission of large amount of data. Also, the success rate of asynchronous transmission is highly depends on the on the recognition of the start and stop bits. The transmission speed of asynchronous communication is slower than that of synchronous communication.

Differentiating synchronous communication from asynchronous communication:

S.NO	Synchronous Transmission	Asynchronous Transmission
1.	Data is transmitted as frames or blocks.	Data is transmitted as characters or bytes.
2.	Transmission rate is faster.	Transmission rate is slower.
3.	It is expensive.	It is economical.
4.	The time interval between blocks is constant.	time interval is random.
5.	It is regulated by an external clock signal also shared with the receiver.	No need of synchronized clocks, as start and end bits are used.

Isochronous Communication: isochronous communication is the hybrid version of both synchronous and asynchronous communication. It inherits the start and stop bit from asynchronous communication i.e. start and stop bits are added to each character before transmission. Characters are transmitted at fix intervals i.e. between twocharacters an exact multiple of one character time interval is added instead of a random time interval, e.g. say t is the transmission time of a character frame with start bit, stop bits and parity bit, then the interval between any two charater frames could be n, where n belongs to positive integer.

2.4 ANALOG AND DIGITAL DATA TRANSMISSION

Data can be transferred in the form of analog signal or digital signal. In data communication, the data is represented in the form of electromagnetic signal like micro ware, radio wave, infrared for wireless, electrical voltage for metallic wires and light for fiber optical cable. ,radiowave, microwave, or infrared signal. Continuous data (like voice, data, image, signal or video) is transmitted through analog signals and called as analog transmission and discrete data is transmitted through digital signals.

Analog Data Transmission: Analog signals are generally represented in the form of sinusoidal wave. Analog signals are continuous in nature i.e. either amplitude or the frequency are continuous in nature and capable to represent continuous data. Video and audio data are transmitted as analog signals.

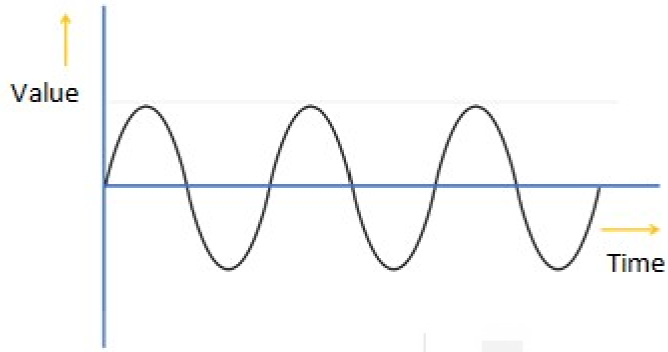

Figure: Analog Signal

As shown in figure the signal value is continuous function of time. Over the distance, signal strength decreases so, to regain the signal strength amplifiers are installed along the transmission line. One of the major issue with analog signals is the accumulation of error during the journey. Signal is distorted during the journey due to external factors and amplifiers are dumb devices which simply increases the strength of input signal carrying noise also. So, along with the data signal, noise level is also regained which is very difficult to separate from the original signal.

Analog signals are not suitable for the transmissions require high level of accuracy. Analog signals can be transformed into digital signals by quantizing or sampling techniques.

Digital Data Transmission: contrast to analog signal being a continuously variable wave form, digital signal is a series of discrete pulses. Digital signals are used to transmit discontinuous data or events.

All the computing devices are based on digital data. Computers generate data in form of digital signals.

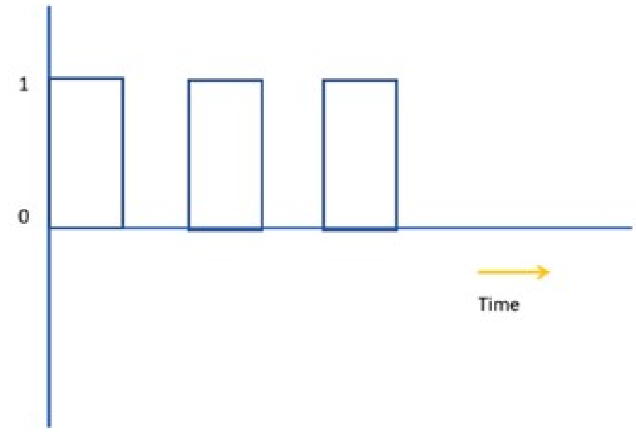

Figure: Digital Signal

As shown in figure above, digital signals are represented in the form of square waves. Digital signals also loss their signal strength or power over the distance. Repeater devices are used to regain the energy level of the signal. Repeater regenerates original level (0 or 1) by measuring the signal values at regular intervals hence the loss of information due to attenuation during transmission does not affect the original signals.

Let us compare analog and digital signals:

Analog	Digital
It is a continuous signal representing real time measurements.	It is dicreteor time separated signal generated by digital modulation.
Sine wave is used to denote the analog signal.	Digital signals represented by square waves
Analog signal can represent continuous range of values.	Discrete values can be represented with digital signal.
Data produced by light sensors, FM radio, temperature sensors, etc are analogin nature and could be represented as analogsignals.	Compact Drives, Digital video Disks and Computers etc. digital signal.
Bandwidthofanalog signal is low	Bandwidth of digital signal is high.

Over the distance analogsignal is deteriorated by noise.	Digital signal is less deteriorated compared to analog signal.
Analog signal is best suitable to transmit video and audio data.	Most of the computing devices works on digital values/signals.
Analog signal is prune to observational errors.	Digital signal is not prune to observational errors.
Analog signal represents the complete range taken by signal.	Digital signal represents the values with finite /limited number.

2.5 TRANSMISSION IMPAIRMENTS

As discussed in presious section, in general analog signals are used to transmit the data because of its simplicity and ease. As the media or channel is not reliable, during the transmission of analog signal, it gets deteriorate i.e. the signal changes its form over the distance leading to change of the data/measurement values represented by this analog signal. During the transmission signal may suffer impairments like: attenuation, delay distortion and noise may occure in signal.

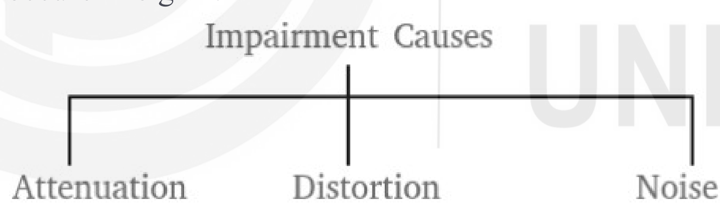

Figure: Causes of transmission inmpairments

2.5.1 Attenuation

The loss in enegy over the distance due to resistance of the medium is termed as attenuation. Hardware device: Amplifiers are installed in between to compensate the losses and regain the signal strength. Attenuation in a signal is measured in decibel and the symbole used for it is dB.Attenuation value is used to compare the signal strengths of different signals. It is also used to evaluate the signal strength of a signal at different positions.

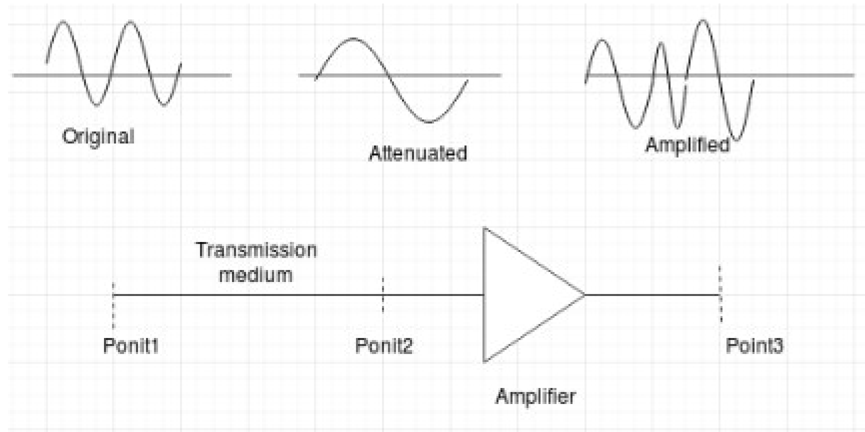

Figure: Effect of attenuation and amplification on a signal

The attenuation or dB of a signal can be calculated as follows:

$$dB(Attenuation) = 10\log_{10}[P_2/P_1]$$

Where $P_1$and P_2 are the powers of the signal at sending end and the position where attenuation is to be compared respectively.

For signals where voltage is considered instead of the power, attenuation is calculated using following formula:

$$dB(Attenuation) = 20\log_{10}(V_2/V_1)$$

Where, V_1 and V_2 are the voltage at he voltage at sending end and the position where attenuation is to be compared respectively.

The reason for attenuation is the resistance of the medium. Some energy is lost to propagate through the medium (wired or wireless) and this loss in energy is termed as attenuation. Attenutaion is less for short distanace and increases with distance. The extent of energy lossdirectly proportional to the frequency i.e. higher attenuationis observed in signals with higher frequencies. An attenuated signal can never be reconstructed to its original form.

2.5.2 Delay Distortion

Distortion is some change in the original. In data communication data is transmitted with a range of frequencies known as the bandwidth of the channel. In general delay distortion is observed in complex or composite signals. A complex/composite signal can be decomposed into many signals of different frequencies. Each frequency component travels with different speed through a medium.Therefore, signals with different frequencies experience different amount of delay during transmission through a channel. This phenomina is usually noticed in fiber optical cable.

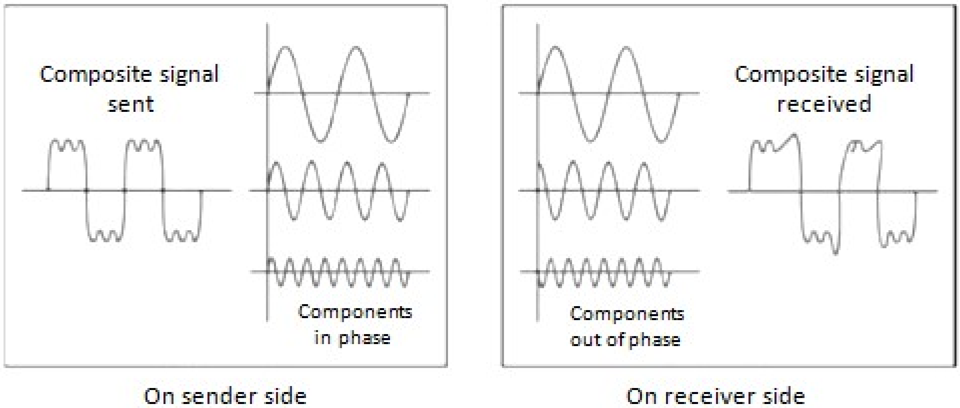

Figure: Delay distortion in a signal

2.5.3 Noise

Noise includes all signals other than the signal transmitted by the source or transmitter. During transmission many types of noise like thermal noise, induced noise, impulse noise and crosstalk noise may mix up with original signal.

Thermal noise: when potential difference is applied across a wire, electrons come into random motion within the wire generating an extra signal, causing thermal noise in the propagating signal.
Induced noise: Sometimes you might have experienced the sound distortion in the speakers when placed into the near vicinity of a mobile and is in use for making a call. This is known as the induced noise.
Impulse noise: sometimes during the transmission the signal may experience energy or voltage spikes. Due to impulse noise in digital signal, sometimes many bits may be lost
Crosstalk noise: it is the interference of one wire with the other. When two wires come into nearby vicinity of each other, signals in one affects the signal of other, it is called as the crosstalk noise. For example, it is often observed that during conversation on telephone, chat of another conversation is heared, it is due to the crosstalk noise between two wires.

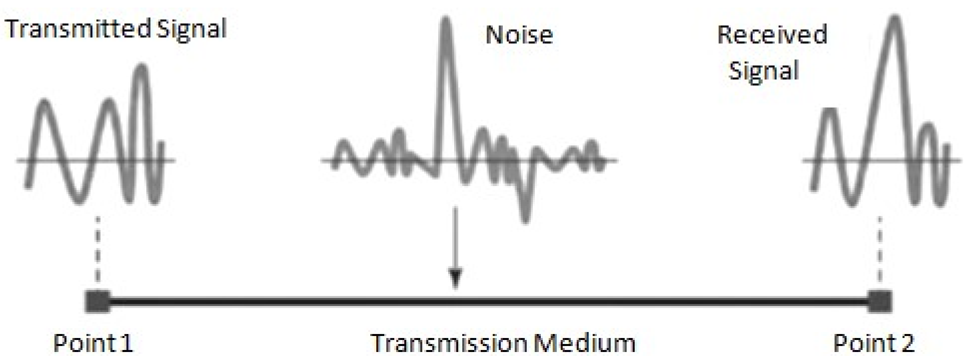

Figure: Noise in transmitted signal

2.5.4 Signal to Noise ratio

Signal to Noise Ratio (SNR) is defined in terms of dB. SNR is used to define the quality of a signal at any point. It is the measure of signal strength at a point along transmission path. It is the ratio of signal power and noise power. Lower the noise, higher the SNR value, and better is the signal quality.

SNR = Avg Signal Power / Avg Noise Power.

SNR = S/N dB

2.5.5 Concept of Delays

The amount of time taken to reach a packet from source to destination is termed as delay. Delay isone of the major factor in networking considered for evaluation of Quality of Service of a system. Delay is considered as performance evaluation parameter of network algorithms, like flow control and routing. A packet has TTL (time to live) value, on expery the packet will be dropped by router, if the delay is more than TTL, the packet will be dropped and transmission is required. Therefor, a large delay is disastrous for data transfer.

In the network a packet may experience delay due to many factors i.e.buffering, link capacity, congestion etc. Some of the delay components are: Transmission delay, Propagation delay, Queuing delay, and Processing delay. Two major delay components: Transmission delay and Propagation delay are discussed here.

Transmission delay: It is the time taken to put the packet on the channel or link. Transmission delay is dependent on the link capacity or the link bandwidth and the data size to be transmitted. Usuallytransmssion delay is caused due to the queuing capability of the intermediate routers. If intermediate routers are using store and forward method of forwarding, transmission delay will be experienced on each intermediate router i.e. transmission delay is not a constant quatity, it is a variable quantity and depends on the traffic load on router as well as the number of routersin the transmission path. Transmission delay can be calculated as:

T_{trans} = Packet size / Bandwith of channel

Propagation delay: Data signal travel through channel over long distances in the Internet, starting from computer to local ISP to satellite to another end ISP station to destination computer. It consumes time to complete this journey and is termed as the propagation delay. Travelling of a signal depends on how easily and with what energy a signal propagates through channel. This delay is experienced in all types of channels and is dependent on physical property of the channel. It is the time duration between: the last bit of packet transmitted at sender and the last bit is received at receiver.

Propagation delay can be calculated as follows:

T_{prop}= Distance between two ends/ Speed of signal in the channel

2.6 TRANSMISSION MEDIA AND ITS CHARACTERISTICS

A medium is required to transmit the data from one place to another. Transmission mediua are classified into two major categories: guided media and unguided media. Guided media like: metal wire and fiber optics guide the signal in which direction to propagate whereas, unguided media like: wireless signals such as radio waves, Inrared etc.

2.6.1 Guided media

Guided media are the channels in which an instrument or an agent or a conduit is provided to transmit the data between devices. A signal traveling through guided media is restricted by the physical boundaries of the channel. Commonly used channels of guided media in networking are twisted-pair cable, coaxial cable, and fiber optical cable.

Twisted pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fiber is a cable that accepts and transports signals in the form of light.
Twisted Pair cable: in twisted pair cables, copper metal is used and responsible to carry the data signal in the form of electic current. It is the oldest transmission channel and widely used in todays scenario also due to easy and simple installation and maintenance and the lower cost.
Twisted pair cable used in networking consists of 4 pairs of copper wires, each pair consisting of 2 wires. The pairs are twisted to overcome the noise induced by electromagnetic effect of nearby similar pairs. The in total 8 wires are provided with fixed color coding, as shown in figure below. Further, twisted pair cable is also divided into two categories: UTP and STP.
UTP is unshielded twisted pair, in which each pair is not shielded separately whereas, in Shielded Twisted Pair (STP) each pair is shielded as shown in fibure below.

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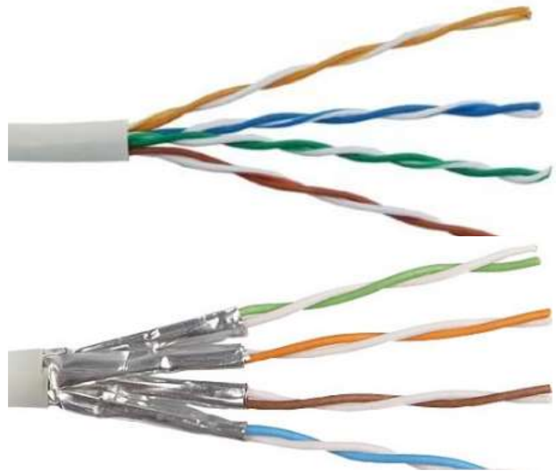


(a) (b)

Figure: Twisted Pair Cable (a) UTP (b) STP.

Both analog and digital transmission is possible through twisted pair cable. The capacity or bandwidth of the channel is defined by the thickness of the metal cable and the distance travelled. The adaptor used forthese cable is termed as RJ45. UTP cable is able to transmit the data upto 100 meters of distance without any repeater or switch. Twisted pair cables are used in Ethernet/LAN technology.

Coaxial Cable: In coaxial cable two conducting wires are used and both are coaxial to each other. Cross section of acoaxial cable is shown in figure below.

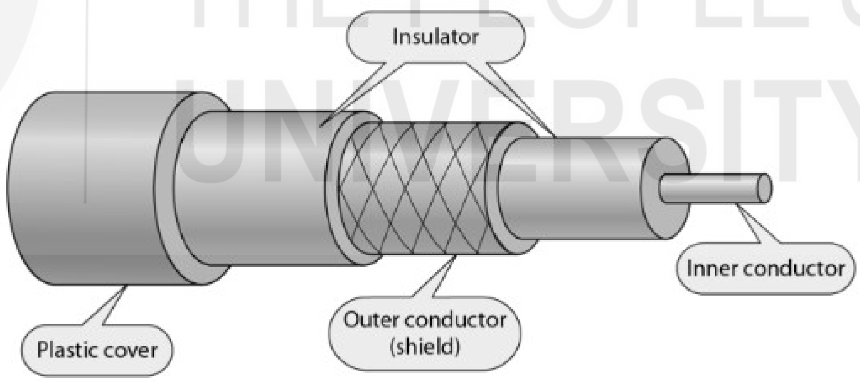


Figure: Coaxial cable

The inner wire also termed as core wire is used for signal transmission is generally of copper metal. The core wire is surrounded by an insulator. Outside the insulator metal net (outer conductor) coaxil with the inner or core wire. The outermost layer is the plastic cover providing the insulation and protection to conductors.
The bandwidth of coaxial cable is more than the twisted pair cable. Connectors used in coaxial cable is BNC (Bayonet Neill-Concelman). In figure widely used

coaxial cable connectors: BNC Connector, the BNC T connector and the BNC terminator are displaced.

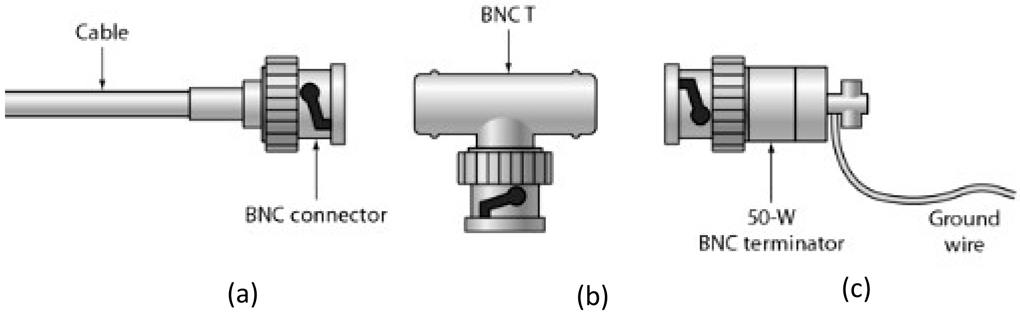


Figure: Coaxial cable BNC connectros

Based on the type of signal a coaxial cable carry, it can be further classified as: Baseband coaxial cable and Broadband coaxial cable.
Baseband coaxial cable is used to transmit digital signals. With baseband coaxial cable data can be transmitted over 1 km at a rate of 1 to 2 Gbps.
Broadband coaxial cable system is used to transmit analog signals over standard television cable network. Broadband term is taken from the telephone system, which means anything wider than 1 MHz. In networking broadband cable is defined as the cable carrying analog transmission.
Fiber Optic Cable: Fiber optic cable works on the total internal reflection principal of light. The data is transmitted in the form of laser light through fiber optics. As shown in figure below, it composed of very thin (even thinner than hair) silicon glass termed as fiber and is coated with a reflective surface (restricts the light signal to escape). Each fiber optic strand can support thousands of speech channels and multiple TV channels. Fiber optic cable provides data transmission at a very high speed. It provides a bigger bandwidth space. One of the biggest advantage of fiber optic cable is its immunity towards electromegnatic interference. The installation and maintenance is costly and require skills.

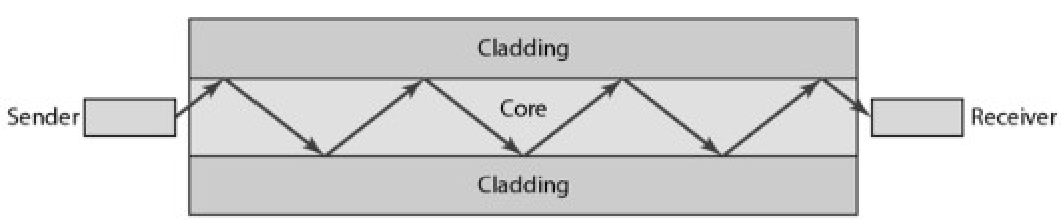


Figure: Fiber optical cable internal view

In fiber optic cable based on its physical characterstics can transmit the data in the form of light in two modes namely: multimode and single mode, where multimode further can be classified as: Step-index and Graded-index.

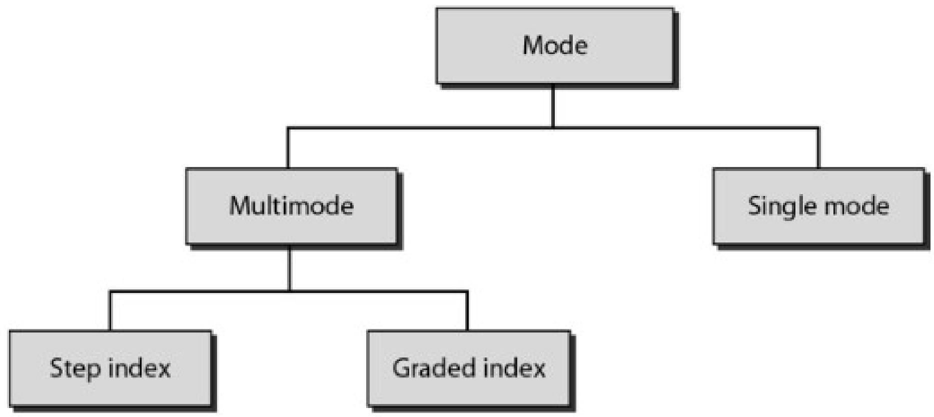


Figure: Fiber optic cable transmission modes

In multimode fiber optic cable multiple light beams or rays can propagate in the core (glass) following different paths.

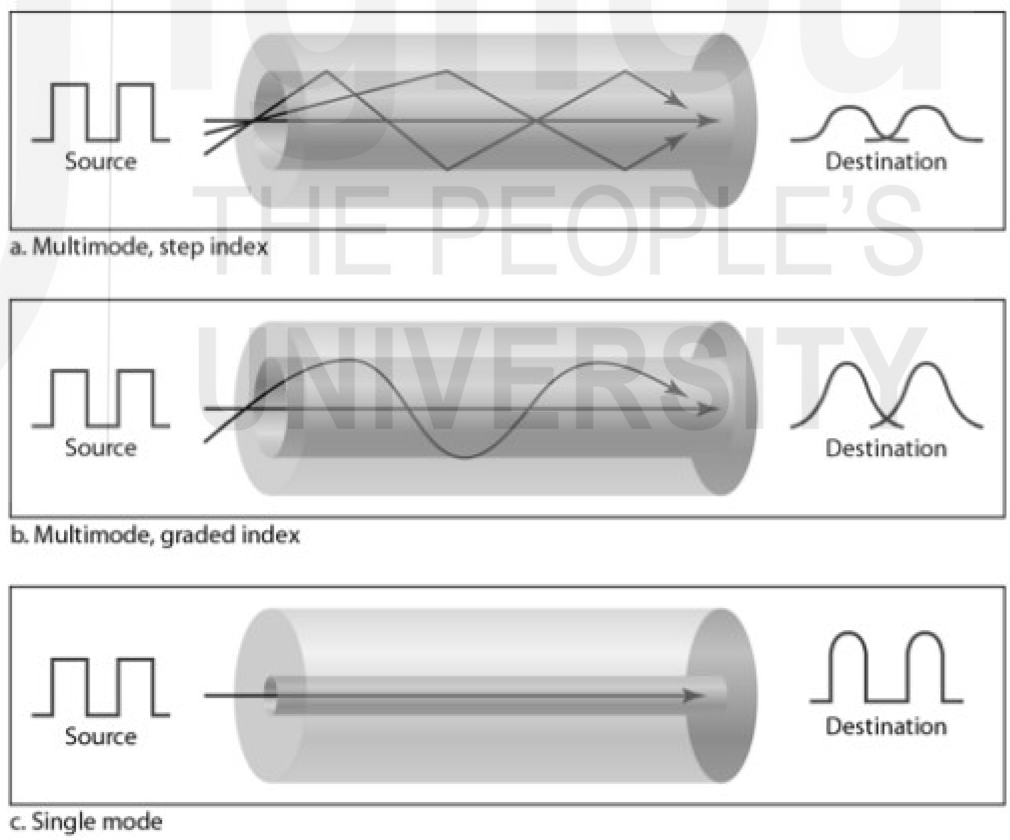


Figure: Light propagation through fiber optic cable

Multimode step index fibre optic cable: as shown in figure above the light signals propagates in straight lines. The core of multimode step index fiber is madeup of constant density throughout the glass fromcentre to edge. In this the light signals face a sudden change in the direction (at the boundaries of glass), which distort the signal.

Multimode graded index fibre:the light signals propagates some how like a wave form. The light signal doesnot face a sudden change in direction which reduces the distortion of signal. The core of multimode graded index fiber is madeup of varying density with highest density value at the centre and decreases progressivelytill edge of the core.

Connector used in fiber optic cable are shown in figure below is SC connectors.

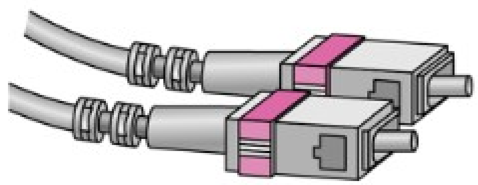


Figure: SC connector for fiber optic cable

Single Mode

In single mode fiber cable the light beam propagates almost horizontal. In single mode fiber cable the core is based on stepindex fibre with very small size diameter as compared to that in multimode fiber. The density of core is very less as compared to multimode core.

2.6.2 Unguided media

This is also termed as unbounded media. There is no instrument or agent or any physical entity to transmit the data through unguided media. The signal is broadcasted through air and is open to access to all, one has a device with capability to receive the signal can have access to data. The data is transmitted in the form of electromagnetic waves through unguided media. Electromagnetic waves transmits through free space in unrestricted manner. The communication through unguided media is in general termed as wireless communication. Data transmission through unguided media can take place in many ways:gound propagation, sky propagation and line-of-sight propagation.

Ground Propagation: Ground propagation takes place through radio waves close to the earth surface. In these type of propagation, radio wave signals with low frequency in general from the spectrum below 2 MHz are used. These signals propagates in all directions termed as omni directional propagation. In this, the signals traverse in the curvature same as the that of the planet.

Figure: Ground Propagation

Sky Propagation: In sky propagation as name suggests, the radio waves with frequencies higher than ground propagation (in the range 2 MHz to 30 MHz) are used to transmit the data. In this, radio waves are emitted towards the sky and after hiting ionosphere are reflected back towards earth. Sky propagation type of transmission is used for sending data over large distances.

Figure: Sky Propagation

Line-of-sight Propagation: Line of sight propagation is used to transmit the signals in a straight line between transmitting and receiving antennas. This type of transmission uses signals of very high frequency in the range above 30 MHz.In line of sight propagation both end (transmitter and the receiver) should be visible to each other that is, no object should be there in between the line of sight of both ends.

Figure: Line of signt propagation

The wireless transmission can be classified as follows:

Radio waves
Micro waves
Infrared waves

2.7 WIRELESS TRANSMISSION

Wireless transmission is one in which data is propagated through air medium (atmospheric space) In wireless transmission the two communicating devices are not required to be connected through any physical channel. Wirelesscommunication is easy to setup/install and maintain. Wireles channel is helpful and convenient to rach locations otherwise difficult to laydown or install the physical or wired channel. In general following wireless media are widely used:

- Microwave
- Radio wave, and
- Infrared.

2.7.1 Microwave Transmission

Microwaves are the electromagnetic waves of freaquencies in the range 1 and 300 GHz. Due to high frequency these narrow focused and unidirectional. As the signal is narrow focused, the both ends transmitter and receiver antennas should be in line of sight to each other. To transmit the signals at long distances the height of antennas need to be very tall. Wicrowave signals cannot paas through obstacles due to its high freqhency band, hence the signals can not be delivered if the receiver is not in open area. As the frequency band is wide with range 1 to 300 GHz, data transmission can take place at very high speed. Many simultaneous transmissions can take place by allocating frequency sub bands separated apart from each other to overcome the interference of each other. Some of the frequency bands are reserved and need to take permission before use from authorities.

As shown in figure below, antennas used to transmit and reeive microwaves are classified into two categories: **Parabolic Dish antenna**and **Hornantenna** .

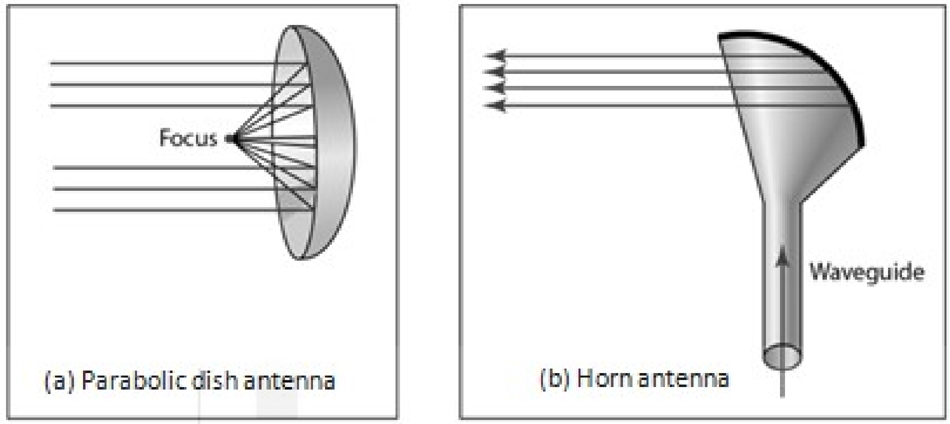


Figure: Microwave antennas

The design of a parabolic antenna is such that a wide range of waves are collected and concentrated to a point forming the high strength signal otherwise would not possible with a single point receiver. The parabolic antennas are good for receiver end.

Aother antenna used in microwave transmission is the horn antenna. The physical shape of horn antenna is like a huge scoop as shown in figure above. The signals transmitted hits the inner surface and transmitted as a narrow parallel beams. At the receiver end the signals hit the scooped shape of the horn, and collected back.

Further the microwave transmission is classified into two categories as:

1. Terrestrial Microwave

   Terrestrial Microwaves are used to communicate the data between transmitter and receiver both located on Earth with lower endfrequency typically between 2-6GHz or 21-23GHz of microwave range. The terrestrial microwave transmission follows line of sight transmission. In terrestrial microwave transmission, parabolic antennas are used, installed in line of sight with each other.
   Environmental factors like rain, fog and wind do not affect the microwave signal transmission Eventhough as the wireless communication is open to all and is vulnerable to electronic eavesdropping.

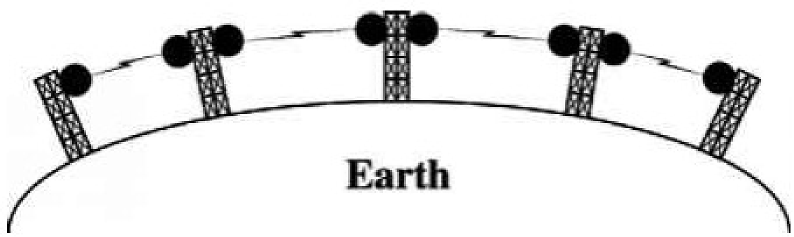


Figure: Terrestrial microwave transmission

2. Satellite Microwave

   Satellite Microwaves are used to trnamsmit the data between the locations not in line of sight to each other i.e. on opposite side of Earth. In this system, depending on the geophysical locations of the two ends the communication may take place between two satellites or beween transmitter on Earth to satellite or beween the satellite or the receiver on the Earth. The parabolic antennas are used on Earth.

   Satellite microwave transmission uses lower end of the frequencies in the microwave frequency band typically in the range 4-6 GHz. Due to lower frequencies environmental factors affect the transmission of satellite microwave signals. Many of the times you might have noticed that in bad weather TV signals become very poor.

2.7.2 Radio Transmission

Radio waves are electromagnetic waves of frequencies in the range 3 KHz and 1 GHz. Radio waves are propagated in omnidirection that is in all directions. Due to propagation of radio waves in omnidirectionthe communicating antennas at both ends need not to be in line-of-sight to each other. Due to omnidirection propagation of radio waves, antennas transmitting and receiving signals with same or nearby frequencies may experience interference i.eunlike microwave where same frequencies can be reused, in radio waves frequency reuse for another transmission is not possible.
Low frequency radio waves are capable to penetrate the walls and can be received even within the building. This advantage some times also becomes disadvantage as signals cannot be restricted so a communication cannot take place in isolation. Radio wave communication is useful in AM and FM radio, TV, maritime radio and cordless phones communication.

2.7.3 Infrared and Millimeter Waves

Infrared waves are used in short range communication limited to some feet. Infrared waves have frequencies in the range 300 GHz- 400 THz. These waves cann'tpenetrate walls making it line-of-sight communication. Infrared waves are generally used in remote control systems. Sun's rays also have infrared waves, which can interfere the other communication making communication in open space impractical.

2.8 WIRELESS LAN

Wireless LAN is a type of local area network in which systems are connected to each other wirelessly and communicate through radio wave channel. The radio waves propagate in omni directional way so, in wireless LAN systems are not to be in line-of-sight with transmitter. Wireless LAN is easy to install and maintain. Wireless LAN is generally installed in places where computers or end devices are located in close proximity like computer lab in school and college etc. It provides a bandwidth of 1 to 2 Mbps.

2.9 SUMMARY

In this unit, we have discussed about the data communication technologies. Starting from what is data communication, we have studied about the basic terminologies used in data communication i.e. channel, baud, bit rate, bandwidth, and frequency etc. Channel is the medium or the path carrying the data signals from source to destination. Baud is the number of symbols transmitted per unit time and the bit rate can be defined as [baud rate x bits in a baud]. The frequency is measured in Hz and is the number of cycles completed by a signal in one second.

Data transmission can take place through two modes: simplex mode and duplex mode. In simplex mode, the data can be transmitted through channel in one direction only. Duplex mode can further classified as: Half duplex mode and Full duplex mode. In half duplex mode, data can be transmitted in both direction but in one direction at a time whereas in full deplex mode, data can be transmitted in both directions simultaneously. Further we have studied various data communication types: parallel communication and serial communication. Serial communication further can be classified as Asynchronous, Synchronous and Isochronous communication. Based on the type of values a signal can represent, signals can be classified into two categories namely: analog signals and digital signals. during he signal propagation through channel, it may get distorted due to which the original signal may experience errors in the form of: attenuation, delay distortion, and noise. The loass of signal strength over distance is termed as attenuation. Delay distortion is due to the varied speed of signal of different frequencies. All unwanted signals in the data signal are considered as noise. A signal strength at any point is measured in terms of SNR (signal to noise ratio) which is the ratio of powers of signal and that of the noise. During transmission of

the signal, itexperiences certain delays like: transmission delay, propagation delay etc. Transmission delay is the time required to put all the data bits to be transmitted on the channel and propagation delay is the time taken by the signal to travel from source to destination in the channel.
Further, two types of transmission media namely: guided and unguided are discussed. Guided media are the physical medias like conductor wires and fiber optic cable whereas, unguided media is the open sky. In general, coaxial cables, twisted pair cables and fiber optic cables are widely used guided media in communication. Microwave Transmission, Radio Transmission, Infrared and Millimeter Waves are types of wireless media. Depending on the distance, security and installation cost one of the wireless channels could be used.

2.10 SOLUTIONS/ANSWERS

Q1. What is Data Communication?
Solution: Exchange of information among devices located over various geographically distant locations is termed as data communication. Data communication systemiscombinationof both softwares and hardwares. The characterstics of an efficient data communication system are: correct, accurate and timely data delivery.

Q2. Explain the term bandwidth and why it is useful?
Solution: Bandwidth is the data carrying capacity of the channel/media per unit time (It can be defined for the bus within a computer or the channel of computer network).
More is the bandwidth higher is the throughput of the medium. Bandwidth of medium is the range of the frequencies used to transmit the data i.e. the difference between highest and the lowest frequencies the medium can carry.
The bandwidth for digital devices is represented as bps (bits per second) whereas, expressed in Hertz (Hz) or number of cycles per second for analog devices.
Q3.Calculate the frequency of a sine wave which completes one cycle in 3 sec?
Solution: Period of the signal (T) = 3 sec
frequency (f) = 1/T
= 1/3 = 0.33Hz.
Q4. Compare Parallel and Serial transmission methods and mention the situation
Solution: where Parallel transmission is a better choice as compared to serial transmission.
Solution:

Parallel Transmission	**Serial Transmission**

In parallel transmission all the data bits are transferred parallel at a time.	In serial transmission data bits are transferred one by one serially.
Data bits are transmitted on multiple wires parallely.	Data bits are transmitted on a single wire serially.
Data transferred at a very high speed.	Data transmission rate is slower than that in parallel transmission.
It is costly and complex system.	It is cheaper and simpler than parallel transmission.
Best for transmitting largeamount data over short distance with high accuracy.	Best suitable for transmitting data over long distances.

Q5. How Full duplex data transmission mode is more challenging than other two modes? Solution: In Full duplex transmission data can be transmitted in both directions simultaneously. In full duplex both ends should be capable of transmission and receiving capabilities.

Q6. Bring out the difference between Synchronous, Asynchronous transmission.
Solutions:

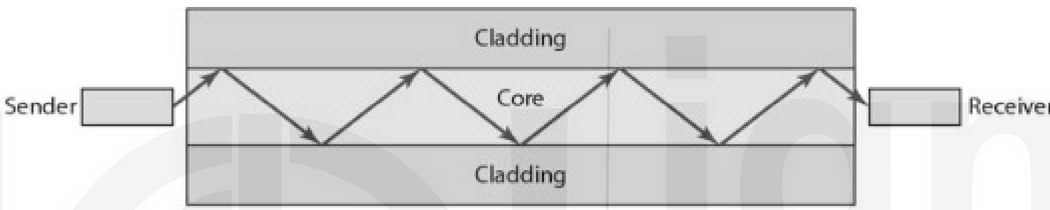
S.NO	Synchronous Transmission	Asynchronous Transmission
1.	Data is transmitted as frames or blocks.	Data is transmitted as characters or bytes.
2.	Transmission rate is faster.	Transmission rate is slower.
3.	It is expensive.	It is economical.
4.	The time interval between blocks is constant.	time interval is random.
5.	It is regulated by an external clock signal also shared with the receiver.	No need of synchronized clocks, as start and end bits are used.

Q7. What is Analog data transmission?
Solution.Analog data transmission takes place through analog signals. Analog

signal are used to transmit continuous data and are represented by sine wave. Audio, video and music etcare types of analog signal.

Q8. Describe the structure of optical fiber and mention its advantages and disadvantages.

Solution. Fiber optic cable works on the total internal reflection principal of light. The data is transmitted in the form of laser light through fiber optics. As shown in figure below, it composed of very thin (even thinner than hair) silicon glass termed as fiber and is coated with a reflective surface (restricts the light signal to escape). Each fiber optic strand can support thousands of speech channels and multiple TV channels. Fiber optic cable provides data transmission at a very high speed. It provides a bigger bandwidth space. One of the biggest advantage of fiber optic cable is its immunity towards electromegnatic interference. The installation and maintenance is costly and require skills.



Q9 Explain the working of Wireless LAN's.

Solution. Wireless LAN is a type of local area network in which systems are connected to each other wirelessly and communicate through radio wave channel. The radio waves propagate in omni directional way so, in wireless LAN systems are not to be in line-of-sight with transmitter. Wireless LAN is easy to install and maintain. Wireless LAN is generally installed in places where computers or end devices are located in close proximity like computer lab in school and college etc. It provides a bandwidth of 1 to 2 Mbps.

References:

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UNIT 3 DATA ENCODING AND MULTIPLEXING

Structure **Page Nos.**

3.0 INTRODUCTION

You might be aware from the description in the previous blocks, how data is transmitted through a channel (wired or wireless) in the form of an analog or digital signal. In this unit, we will elaborate on the techniques to produce these types of signals from analog or digital data. We will look at, in brief, on the following encoding techniques [*Figure1*]:

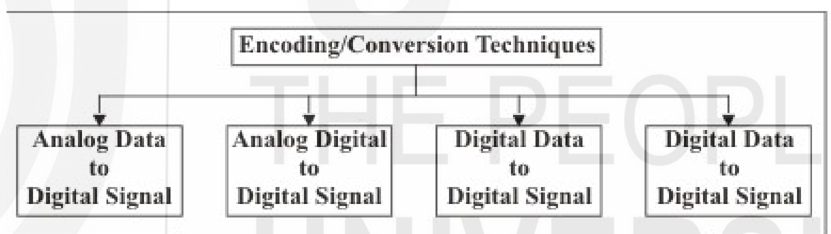

Figure 1: Encoding/Conversion Techniques

- Analog data as an analog signal
- Analog data as a digital signal
- Digital data as an analog signal
- Digital data as a digital signal

We will confine ourselves to, a treatment of the most useful and common techniques only, and will skip over the many other kinds of encoding of data that can be performed.

3.1 OBJECTIVES

After going through this unit, you should be able to describe:

- what encoding means;
- why encoding is needed;
- some different types of encoding;
- mechanisms and techniques used for encoding, and
- when each type of encoding is used with examples.

There are several other types of encoding as well as encoding techniques that will not be discussed in this unit.

3.2 ENCODING

Usually, we cannot send a signal containing information directly over a transmission medium, at least not for long distances. For example, the sound of our voice can only travel a few hundred meters. If we want to send our voice to the next city, we have to be able to transform our voice, to, an electrical signal that can then, be sent that long distance. Then, we would also need to perform the reverse conversion at the other end.

So, we see that, to send a signal over a physical medium, we need to encode or transform the signal in some way so that the transmission medium can transmit it. The sender and the receiver must agree on what kind of transformation has been done so that, it can be reversed at the other end and the actual signal or information can be recovered. The information content in a signal is based upon having some changes in it. For example, if we just send a pure sine wave, where the voltage varies with time as a sine function, then, we cannot convey much information apart from the fact that there is some entity at the other end that is doing the transmitting.

The information content of a signal is dependent on the variation in the signal. After sending one cycle of a sine wave there is no point in sending another cycle because we will not be able to convey anything more than what we have already done. What is needed is to modify the sine wave in some way to carry the information we want to convey. This is called modulation.

You know that there are two kinds of signals : analog and digital. Likewise, there are two types of information that we may want to transmit, again analog and digital. So, we get four basic combinations of encoding that we may need to perform, depending on the signal type as well as the information type. We shall look at each of these four types briefly in this unit and will also study the ways in which the actual encoding can be done.

3.3 ANALOG-TO-ANALOG MODULATION

Let us first look at a situation where our signal and information are both of the analog type. The act of changing or encoding the information in the signal is known as **modulation.** A good example of such encoding is radio broadcasting. Here we primarily want to send sound in some form over the atmosphere to the receiving radio sets. The sound is converted into an analog electrical signal at the source and is used to encode the signal which is the base frequency at which the transmission is being done. The reverse process is performed at the radio set to recover the information in electrical form, which is then converted back to sound so that, we hear what was being said at the radio station.

Let us, formulate another definition of modulation. When a low frequency information signal is encoded over a higher frequency signal, it is called modulation [Ref.3]. The encoding can be done by varying amplitude (strength of a signal), period (amount of time, in seconds, a signal needs to complete a cycle) and frequency (number of cycles per second) and phase (position of waveform relative to zero).

Notice the simulations of the word modulation to the word modifying. For instance, an audio signal (one that is audible to human ear) can be used to modify an RF (Radio Frequency) carrier, when the amplitude of the RF is varied according to the changes inthe amplitude of the audio, it is called amplitude modulation (AM), and when the frequency is varied, it is called FM (frequency modulation).

Although, it can be dangerous to use analogies as they can be carried too far, we can understand this process with an example. Suppose we want to send a piece of paper with something written on it to a person a hundred meters away. We will not be ableto throw the paper that far. So, we can wrap it around a small stone and then throw the stone towards the person. He can then unwrap the paper and read the message.
Here, the stone is the signal and the piece of paper is the information that is used to change the stone. Just as sending the stone alone would not have conveyed much to our friend, similarly sending only the base signal (also called carrier signal) will notconvey much information to the recipient.

Therefore, there are three different ways in which this encoding of the analog signalwith analog information is performed. These methods are:

- **Amplitude modulation** (AM), where the amplitude of the signal is changeddepending on the information to be sent (*Figure 2 (a)*).

- **Frequency modulation**, where the frequency of the signal is changeddepending on the information to be sent (*Figure 2 (b))*.

- **Phase modulation,** where it is the phase of the signal that is changed accordingto the information to be sent.

Amplitude Modulation

Now, let us go into the details. In this type of modulation the frequency and phase ofthe carrier or base signal are not altered. Only the amplitude changes and we can seethat the information is contained in the envelope of the carrier signal. It can be proud/demonstrated that, the bandwidth of the composite signal is twice that of the highest frequency in the information signal that modulates the carrier.

By international agreement, for radio transmission using AM, 9 KHz is allowed as thebandwidth. So, if a radio station transmits at 618 KHz, the next station can only transmit at 609 or at 627 KHz. It will be clear that the highest audio frequency that can be carried over AM radio is thus 4.5 KHz, which is sufficient for most voice and musical pieces. Actually, the modulating signal is centred around the carrier frequency and extends the composite signal both ways in equal measure. Each of these is called a sideband and therefore, AM radio transmission is dual sideband. This is actually wasteful because the spectrum is unnecessarily used up, but the cost and complexity associated with eliminating one sideband, and performing single sideband AM modulation, have led to the technique not being used widely.

AM radio transmission has been assigned the frequency range of 530 to 1700 KHz. The quality of AM transmission is not very good as noise in the channel (the atmosphere, here) can easily creep into the signal and alter its amplitude. So, the receiving end will likely find a fair amount of noise in the signal, particularly at higher, short wave frequencies. That is why you often get a lot of static in short waveor even medium wave broadcasts.

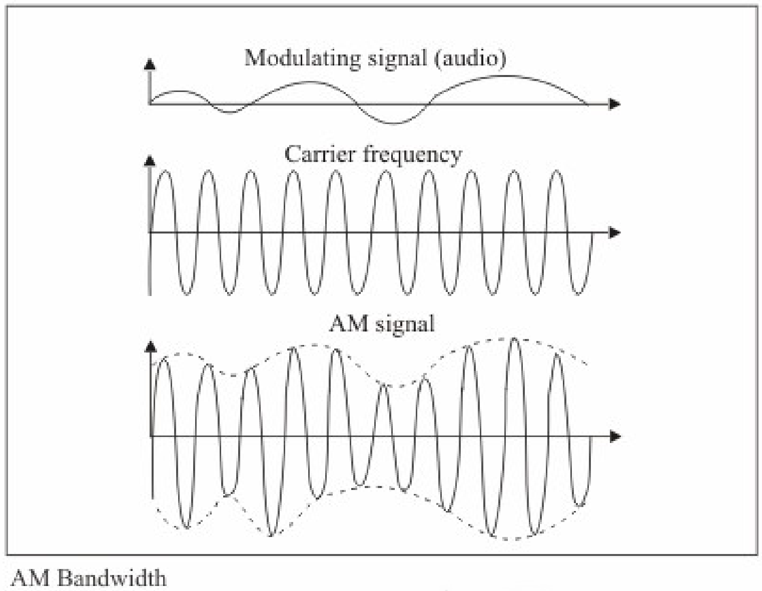

Figure 2 (a): Amplitude modulation

Frequency Modulation

In contrast to Amplitude Modulation, here it is the frequency of the base signal that is altered depending on the information that is to be sent. The amplitude and phase of the carrier signal are not changed at all. It can be shown that this results in a bandwidth requirement of 10 times the modulating signal, centred on the carrier frequency.

This method is the less susceptible to noise and gives the best performance of all data encoding types as far as the quality of the transmitted signal is concerned. Although digital encoding methods may give better performance over multiple hops (because in their case, the original signal can be accurately reconstructed at each hop), FrequencyModulation (FM) is the best as far as single hop transmission goes.

FM radio transmission has been allocated the spectrum range 88 MHz to 108 MHz. As a good stereo sound signal needs 15 KHz of bandwidth, this means that FM transmission has a bandwidth of 150 KHz. To be safe from interference from neighbouring stations, 200 KHz is the minimum separation needed. As a further safety measure, in any given area, only alternate stations are allowed and the neighbouring frequencies are kept empty. This is practical because FM transmission is line of sight (because of the carrier frequency) and so, beyond a range of about 100Km or so, the signal cannot be received directly (this distance depends on various factors). We can therefore, have at most 50 FM stereo radio stations in a given geographical area.

As in the case of AM transmission (having 2 sub bands), FM also has multiple sidebands maybe two or more, which is really wasteful but is tolerated in the interestsof simplicity and cost of the equipment.

In television transmission the bandwidth needed is 4.5 MHz in each sideband. The video signal is sent using amplitude modulation. The audio signal goes at a frequency that is 5.5 MHz over the video signal and is sent as a frequency modulated signal. To reduce the bandwidth requirement, the video signal is sent using a method called vestigial sideband that needs only 6 MHz, instead of the 9 MHz that would have been needed for dual sideband transmission, though, it is more than the 4.5 MHz that a puresingle sideband transmission would have needed.

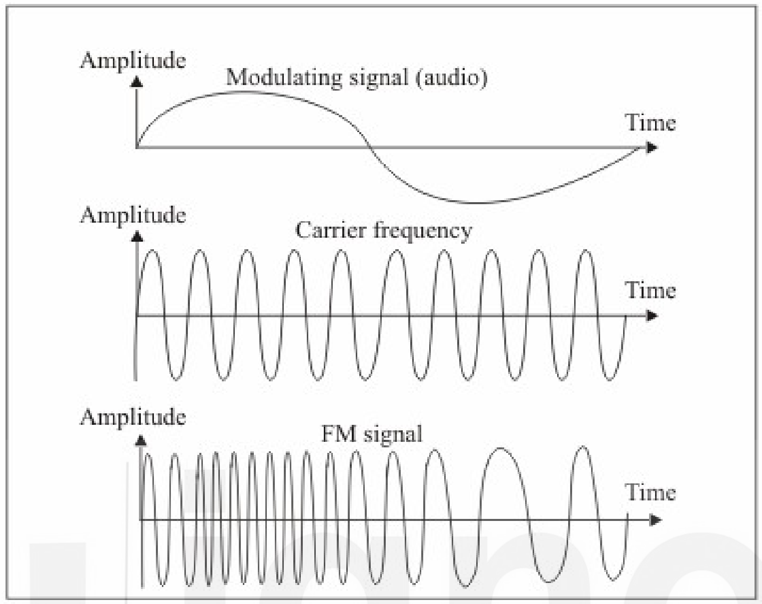

Figure 2 (b): Frequency modulation

Phase Modulation

As you would have guessed by now, in Phase Modulation (PM) the modulating signalleaves the frequency and amplitude of the carrier signal unchanged but alters its phase. The characteristics of this encoding technique are similar to FM, but the advantage is that of reduced complexity of equipment. However, this method is not used in commercial broadcasting.

☞ Check Your Progress 1

1) Why do we need modulation? Would it be right to simply send the informationas the signal itself ?

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2) Why is it that there are at most 50 FM radio stations in a particular geographicalarea? Justify with calculations.

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3) Why is AM the most susceptible to noise, of the three types of modulation?

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3.4 ANALOG TO DIGITAL MODULATION

Natural phenomena are analog in nature and can take on any of the potentially infinite number of values. For example, the actual frequencies contained in a sound made by a human is an analog signal, as is the amplitude or loudness of the sound. One example of coding analog data in digital form is when, we want to record sound on digital media such as a DVD or in other forms such as MP3 or as a ".wav" file. Here, the analog signal, that is, the musical composition or the human voice is encoded in digital form. This is an example of analog to digital encoding.

While, in the case of analog to analog encoding, the motivation was to be able to transmit the signal for long distances, here, the main reason is to be able to change the information itself to digital form. That digital signal can then be transmitted, if necessary, by any suitable method.

One method of encoding is what is called Pulse Code Modulation (PCM). This gives very good quality and when used in conjunction with error correction techniques, can be regenerated at every hop on the way to its final destination. This is the big advantage of digital signals. At every hop, errors can be removed and the original signal reconstructed almost exactly, barring the few errors that could not be corrected.

The first step in PCM is, to convert the analog signal into a series of pulses (*Figure 3*). This is called Pulse Amplitude Modulation (PAM). To do this the analog signal is sampled at fixed intervals and the amplitude at each sample decides the amplitude of that pulse. You can see that at this step, the resulting signal is still actually an analog signal because the pulse can have any amplitude, equal to the amplitude of the original signal at the time the sample was taken. In PAM, the sampled value is held for a small time so that the pulse has a finite width. In the original signal the value occurs only for the instant at which it was sampled.

One question that arises here is, how many samples do we have to take? We would not want to take too many samples as that would be wasteful. At the same time, if, we take too few samples, we may not be able to reconstruct the original signal properly. The answer comes from Nyqist's theorem, which states that, to be able to get back the original signal, we must sample at a rate, that is, at least twice that of the highest frequency contained in the original signal.

Let us do some calculations to see what this means in practice. In the next unit, you will see that a voice conversation over a telephone has a range of frequencies from 300 Hz to 3300 Hz. Because the highest frequency is 3300 Hz, we need at least 6600 samples a second to digitise the voice to be sent over a phone line. For safety we take 8000 samples per second, corresponding to a sampling interval of 125 microseconds.

The next stage in the digitization of the signal is quantisation. In this process, the analog values of the sampled pulses are quantised, that is, converted into discrete values. For example, if the original analog signal had an amplitude ranging from +5v to –5v, the result of PAM might be to produce pulses of the following amplitudes (each sample is taken at say, intervals of 125 microseconds)

+4.32, +1.78, -3.19, -4.07, -2.56, +0.08 and so on.

In quantisation, we may decide to give the amplitude 256 discrete values. So, they will range from –127 to +128. The actual amplitude of +5v to –5v has to be represented in this range, which means each value is of 39 mV nearly. By this token, the first value of +4.32v lies between 110 and 111, and we can decide that such a value will be taken as 110. This is quantisation. The analog values now become thefollowing quantised values:

110, 45, -81, -104, -65, 2 and so on.

The above discrete values are now represented as 8 binary digits with, 1 bit giving the sign while the other 7 bits represent the value of the sample.

In the final stage of the encoding, the binary digits are then transformed into a digital signal using any of the digital to digital encoding mechanisms discussed later in this unit. This digital signal is now the representation of the original analog signal.

PCM is commonly used to transmit voice signals over a telephone line. It gives goodvoice quality but, is a fairly complex method of encoding the signal. There is a simpler method that can be used, but we have to pay the price in terms of the lower quality of the encoded signal. This is called Delta modulation. Here, samples are taken as described above, but instead of quantising them into 256 or more levels, onlythe direction of change is retained. If the sample is larger in value than the previous one, it is considered a 1 while otherwise it is considered a 0 bit. So the sequence above would be encoded as:

1, 0, 0, 0, 1, 1 and so on.

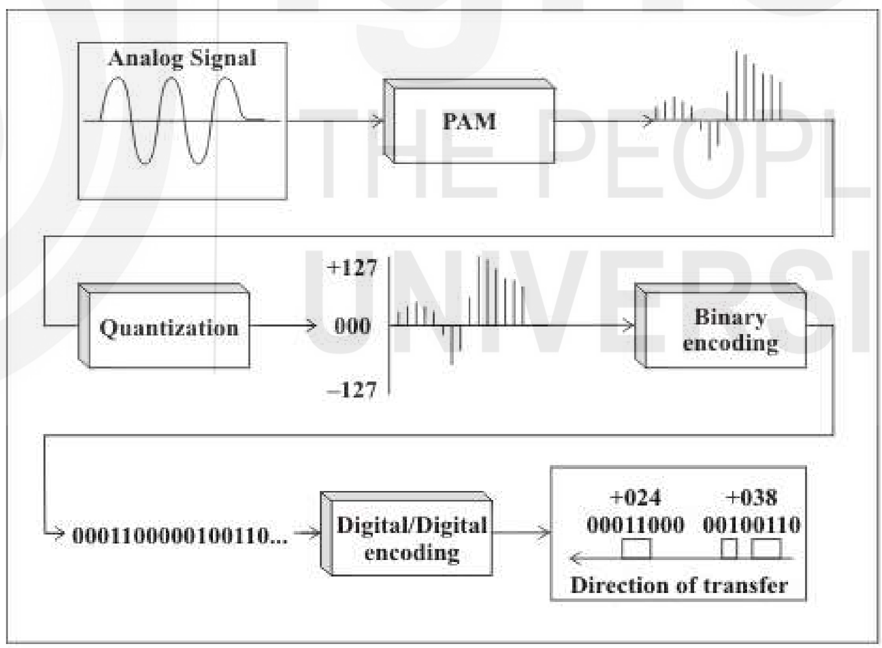

Figure 3: Pulse code modulation

☞ Check Your Progress 2

1) Why is PAM a necessary pre-requisite to PCM? Why can we not use PCMdirectly ?

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2) What would be the minimum sampling interval needed for reconstructing a signal where the highest frequency is 1 KHz ?

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3) Consider a signal where the amplitude varies from +6.4v to –6.4v. If we want to quantise it into 64 levels, what would be the quantised values correspondingto signals of –3.6v and +0.88v ?

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3.5 DIGITAL TO ANALOG MODULATION

So far, we have looked at the codification of analog data, where the source information could take up any value. We also have need for the reverse type of encoding, where binary digital values have to be encoded into an analog signal. This may seem simple because after all, we only have to give two distinct values to the analog signal, but here you need to realise that, we have to be able to distinguish between successive digital values that are the same, say three 1's in succession or two0's in succession. These should not get interpreted as a single 1 or a single 0! This sort of problem did not exist in the case of analog to analog encoding.

One example where such encoding is needed is, when we need to transmit between computers over a telephone line. The computers generate digital data, but the telephone system is analog. So, this situation is a bit different from the transmissionof voice over the telephone, where the source signal is also analog.

In this kind of encoding, one important factor of interest is the rate at which we can transfer bits. Clearly, there will be some kind of limit to the amount of information that we can send in a given time. This depends on the available bandwidth of the carrier signal, that is, the variation that we can permit around the nominal frequencyas well as on the specific technique of encoding that we use.

Here it is important to understand two aspects of the rate of transfer of information. The first is the bit rate, that tells us the number of individual bits that are transmitted per second. The second, and more important from the transmission angle, is the baudrate. This is the number of signal transitions per second that are needed to represent those bits. For a binary system, the two are equal. In general, the bit rate cannot be lower than the baud rate, as there has to be at least one signal transition to represent abit.

We have seen in analog to analog encoding that, the source signal modulates the baseor carrier signal to convey the information that we want to transmit. A similar principle is also used in digital to analog encoding. The carrier signal has to be modulated, but this time by the digital signal, to produce the composite signal that is transmitted. However, while in the analog case there were a potentially infinite number of values that could have been transmitted, here we have only two values (a 0 or a 1) that will modulate the signal.

If, there is no modulation, the carrier signal will be a pure sine wave at the frequency of transmission. You will realise that the baud rate that can be supported also depends on the quality of the line and the amount of random noise that is present in the medium. Like in analog-to-analog encoding, there are three properties of the sine wave that we can alter to convey information. These are the amplitude, the frequency and the phase. In addition, we can also use a combination of amplitude and phase change to encode information more efficiently. Modulation is also referred to as shift keying in this type of encoding technique.

There are three types of digital to analog modulation (*Figure 4*)

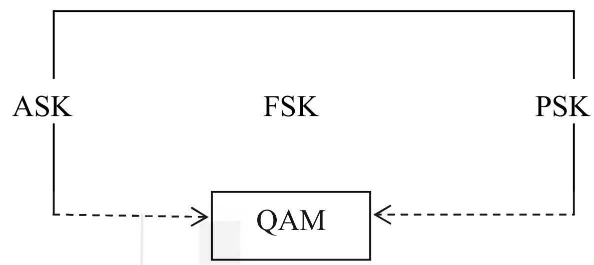


Figure 4: Digital to analog modulation techniques

Now we will take up each one individually.

Amplitude Shift Keying (ASK)

In this kind of encoding, abbreviated as ASK, the amplitude of the analog signal is changed to represent the bits, while the phase and the frequency are kept the same. The bigger the difference between the two values, the lower the possibility of noise causing errors. There is no restriction, however, on what voltage should represent a 1or a 0, and this is left to the designers. For the duration of a bit, the voltage should remain constant.

One method could be to represent the high voltage as a 1 and the negative high voltage as a 0, although the reverse convention could also be used. Sometimes, to reduce the energy needed for transmission, we can choose to represent a 0 or a 1 by azero voltage, and the other bit by the high voltage. This scheme is called on-off-keying (OOK).

Like amplitude modulation in the analog case, ASK is vulnerable to noise. Amplitudechanges easily as a result of the noise in the line, and if, the quantum of noise is great enough, it can result in an error. This would mean that a 0 gets interpreted as a 1 at the receiving end, or vice versa. Noise can occur as spikes or as a constant voltage that shifts the amplitude of the received signal. Even thermal noise, that is fairly constant, can cause a 0 to be interpreted as a 1 or the other way round.

To find the bandwidth requirement of an ASK encoded signal on an ideal, noiseless, line, we have to do a Fourier analysis of the composite signal. This yields harmonics on either side of the carrier frequency, with frequencies that are shifted by an odd multiple of half the baud rate. As most of the energy is concentrated in the first harmonic, we can ignore the others. It can then be shown that the bandwidth requiredfor ASK encoding is equal to the baud rate of the signal. So if we want to transfer data at 256 Kbps, we need a bandwidth of at least 256 KHz.

In a practical case, there will be noise in the line and this ideal cannot be achieved. This increases the bandwidth requirement by a factor that depends on the line qualityand the carrier amplitude. ASK is a simple encoding method but is not commonly used by itself because of efficiency and quality issues.

Frequency Shift Keying (FSK)

Just as in the case of analog to analog encoding, we can encode a digital signal by changing the frequency of the analog carrier. This is frequency shift keying (FSK). When a bit is encoded, the frequency changes and remains constant for a particular duration. The phase and the amplitude of the carrier are unaltered by FSK. Just as inFM, noise in the line has little effect on the frequency and so, this method is less susceptible to noise.

It is possible to show that an FSK signal can be analysed as the sum of two ASK signals. One of these is centred around the frequency used for a 0 bit and the other for the frequency used for a 1 bit. From this, it is easy to show that the bandwidth needed for an FSK signal is equal to the sum of the baud rate and the difference between the two frequencies. Thus, the requirement of FSK is a higher bandwidth that helps us to avoid the noise problems of ASK.

FSK is not much used in practice because of the need for higher bandwidth and thecomparative complex requirement for changing the frequency.

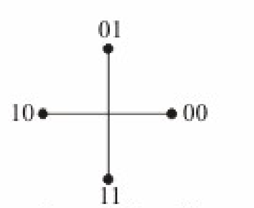
Phase Shift Keying (PSK)

We can also encode by varying the phase of the carrier signal. This is called phase shift keying (PSK) and here, the frequency and amplitude of the carrier are not altered. To send a 1 we could use a phase of 0 while we could change it to 180 degrees to represent a 0. Such an arrangement is not affected by the noise in the line,because that affects amplitude rather than the phase of the signal.

This quality of PSK can be used to achieve more efficient encoding. For example, instead of having only 2 phases, we could have four phases, 0, 90 degrees, 180 degrees and 270 degrees. Each of these phase shifts could represent 2 bits at one go, say, the combinations 00, 01, 10 and 11 respectively. Such a scheme is called 4-PSK. The concept can be extended to higher levels and we could have 8-PSK to send groups of 3 bits in one go. This method can be extended further, but at some point the sensitivity of the communication equipment will not be enough to detect the small phase changes and we are then limited for that reason.

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)



Constellation diagram

Figure 5(a) : 4 PSK

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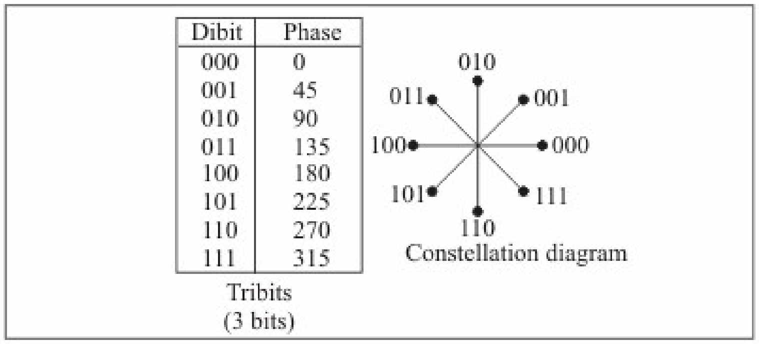

Figure 5(b): 8 PSK

Clearly, the scheme is more efficient than ASK because, we can now achieve a higherbit rate from the same bandwidth. It can be shown that the bandwidth required in 2- PSK is the same as that needed in ASK. But the same bandwidth at 8-PSK can transmit thrice the number of bits at the same baud rate.

Why can we not apply the same trick in ASK? We could have 4, 8 or more amplitude levels to transmit 2, 3 or more bits in one signal transition. The reason is that, ASK is vulnerable to noise and that makes it unsuitable for using many different amplitude levels for encoding. Similarly, FSK has higher bandwidth requirements and so we do not use this kind of technique there as, there is not much to be gained.

Quadrature Amplitude Modulation (QAM)

To further improve the efficiency of the transmission, we can consider combining different kinds of encoding. While, at least one them has to be held constant, there is no reason to alter only one of them. Because of its bandwidth requirements, we would not bother to combine FSK with any other method. But it can be quite fruitful to combine ASK and PSK together and reap the advantages. This technique is called Quadrature Amplitude Modulation (QAM).

We have already seen how ASK is vulnerable to noise. That is why the number of different amplitude levels is small while there may be more phase shift levels possible. There are a large number of schemes possible in the QAM method. For example 1 amplitude and 4 phases is called 4-QAM. It is the same as 4-PSK because there is no change in the amplitude. With 2 amplitudes and 4 phases we can send 3 bits at a time and so this scheme is called 8-QAM (*Figure 6*).

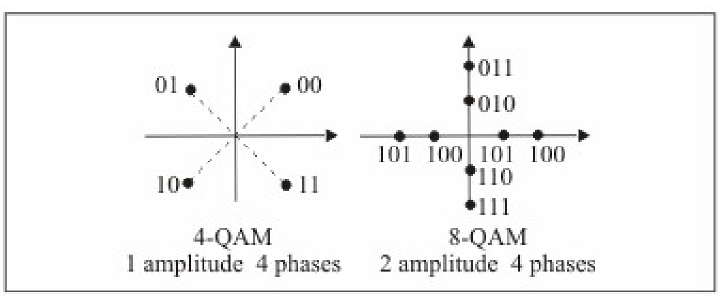

Figure 6: 4-QAM and 8-QAM constellations

In practical application we use 3 amplitudes and 12 phases or 4 amplitudes and 8 phases. Although, this way, we should be able to send 4 bits at a time (this is 32-QAM), to guard against errors, adjacent combinations are left unused. So, there is a smaller possibility of noise affecting the amplitude and causing problems with the signal received. As a further safeguard, in some schemes, certain amplitudes and phases are coupled, so that, some amount of error correction is possible at the physicallayer itself.

Note, that the errors we have talked of here are at the physical layer. By using the appropriate techniques, at the data link layer we achieve error free transmission irrespective of the underlying mechanisms at the physical layer. That is not something that will be looked at in this unit.

It can be shown that QAM requires the same bandwidth as ASK or PSK. So, we are able to achieve a much higher bit rate using the same baud rate. That is why QAM is the method of encoding used currently in data communication applications.

☞ **Check Your Progress 3**

1) What is the difference between the bit rate and the baud rate? Explain with an example.

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2) If our equipment is able to reliably detect differences of voltage of up to 0.5 v, and the probability is 99.74% that noise will cause a change in the amplitude ofour signal of no more than 0.2 v, then how many amplitude levels can one haveif the peak signal strength is to vary from between +3v to –3v ?

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3) If our equipment can reliably detect phase changes of up to 20^{O}, how many phase levels can we use in a PSK system ?

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4) Given the characteristics (of questions 2 and 3 above) and using only half the possible combinations for safety, what level of QAM can be achieved ?

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3.6 DIGITAL TO DIGITAL ENCODING

Let us, now look at the last combination of encoding possible, from digital to digital signals. A simple example is when sending data from a computer to a printer, both of which understand digital signals. The transmission has to be for short distances over the printer cable and occurs as a series of digital pulses. Another example of such encoding is for transmission over local area networks such as an Ethernet, where the communication is between computer and computer.

This kind of encoding is really of three types and we will look at each of the types here. Again, there can be many other mechanisms other than the ones discussed in this unit, but we will look at only the methods that are used in data communication applications. These types are unipolar, polar and bipolar techniques.

Unipolar Encoding

This is a simple mechanism but is rarely used in practice because of the inherent problems of the method. As the name implies, there is only one polarity used in the transmission of the pulses. A positive pulse may be taken as a 1 and a zero voltage can be taken as 0, or the other way round depending on the convention we are using (*Figure 7*). Unipolar encoding uses less energy and also has a low cost.

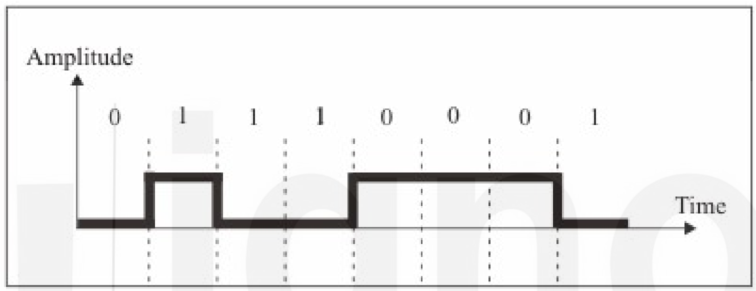

Figure 7: Unipolar encoding

What then are the problems that preclude widespread use of this technique? The two problems are those of synchronisation and of the direct current (DC) component in thesignal.

In any digital communication system the transmitter and receiver have to be synchronised or all information may be lost. For example, in determining when to consider the next bit to have started, a very useful mechanism is of using signal transition to indicate start of the next bit. But, if there is a string of bits of the same value, then this method does not work.

In such a case, we have to rely on the clock of the receiver to determine when a bit hasended. This has potential problems because sometimes there can be delays at the transmission end which may stretch the time of transmission, causing the receiver to wrongly believe that there are more bits than is actually the case. Moreover, drift in the clock of the receiver can also lead to difficulties. So, we cannot directly rely on the transmitter and receiver clocks being accurate enough to achieve synchronisation. One way out of this is, to send a separate timing pulse along with the signal on a separate cable, but this increases the cost of the scheme.

Another problem with unipolar encoding is that the average voltage of the signal is non-zero. This causes a direct current component to exist in the signal. This cannot be transmitted through some components of the circuit, causing difficulties in reception.

Polar

Unlike unipolar schemes, the polar methods use both a positive as well as a negative voltage level to represent the bits. So, a positive voltage may represent a 1 and a negative voltage may represent a 0, or the other way round. Because both positive and negative voltages are present, the average voltage is much lower than in the unipolar case, mitigating the problem of the DC component in the signal. Here, we will look at three popular polar encoding schemes.

Non-return to Zero (NRZ) is a polar encoding scheme which uses a positive as well asa negative voltage to represent the bits. A zero voltage will indicate the absence of any communication at the time. Here again there are two variants based on level and inversion. In NRZ-L (for level) encoding, it is the levels themselves that indicate the bits, for example a positive voltage for a 1 and a negative voltage for a 0. On the other hand, in NRZ-I (inversion) encoding, an inversion of the current level indicates a 1, while a continuation of the level indicates a 0.

Regarding synchronisation, NRZ-L is not well placed to handle it if, there is a string of 0's or 1's in the transmission. But NRZ-I is better off here as a 1 is signaled by the inversion of the voltage. So every time a 1 occurs, the voltage is inverted and we know the exact start of the bit, allowing the clock to be synchronised with the transmitter. However, this advantage does not extend to the case where there is a string of 1's. We are still vulnerable to losing synchronisation in such a case.

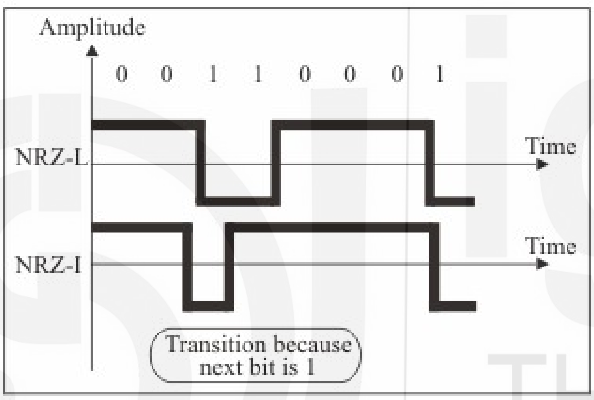

Figure 8: NRZ - L and NRZ I encoding

Return to Zero (RZ) is a method that allows for synchronisation after each bit. This is achieved by going to the zero level midway through each bit. So a 1 bit could be represented by the positive to zero transition and a 0 bit by a negative to zero transition. At each transition to zero, the clocks of the transmitter and receiver can be synchronised. Of course, the price one has to pay for this is a higher bandwidth requirement.

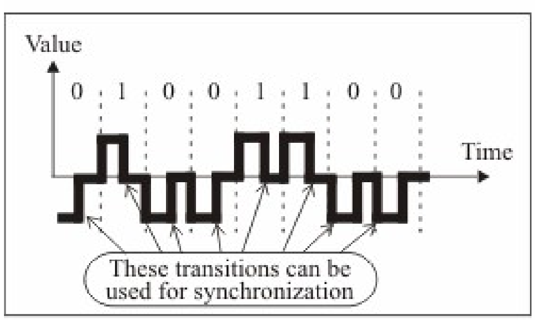

Figure 9: RZ encoding

A third polar method is biphase.encoding. While in RZ, the signal goes to zero midway through each bit, in biphase it goes to the opposite polarity midway through each bit. These transitions help in synchronisation, as you may have guessed. Again this has two flavours called Manchester encoding and Differential Manchester encoding. In the former, there is a transition at the middle of each bit that achieves synchronisation. A 0 is represented by a negative to positive transition while a 1 is represented by the opposite. In the Differential Manchester method, the transition halfway through the bit is used for synchronisation as usual, but a further transition atthe beginning of the bit represents the bit itself. There is no transition for a 1 and there is a transition for a 0, so that a 0 is actually represented by two transitions.

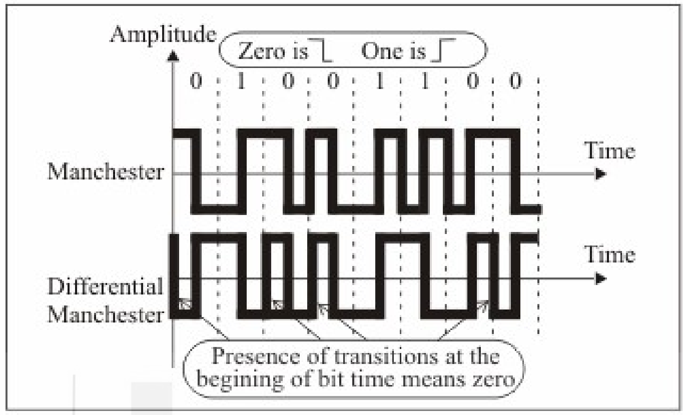

Figure 10: Manchester and differential manchester encoding

Bipolar

This encoding method uses three levels, with a zero voltage representing a 0 and a 1 being represented by positive and negative voltages respectively. This way the DC component is done away with because, every 1 cancels the DC component introducedby the previous 1. This kind of transmission also ensures that every 1 bit is synchronised. This simplest bipolar method is called Alternate Mark Inversion (AMI). The problem with Bipolar AMI is that synchronisation can be lost in a long string of 0's.

One way of synchronising 0's is to use the High Density Bipolar 3 (HDB3) method. This changes the pattern after every four consecutive 0's. The change is to represent the 0 by a positive or negative signal instead of the zero voltage. How do we prevent this from being wrongly interpreted as a 1? It is done by using the same voltage level as was used for the previous 1. As in bipolar encoding, a real 1 bit is represented by the opposite voltage level every time, there is no possibility of confusing the 0 with a
1. So we deliberately violate the convention to achieve synchronisation, a scheme which works because, the violation can be logically detected and interpreted for synchronisation.

HDB3 changes the pattern of four 0's depending on the number of 1's since the last such substitution and the polarity of the last 1. If the number of 1's since the last substitution is odd, the pattern is violated at the fourth 0. So if the previous 1 was positive, the 0 is also made positive to avoid confusion with a 1. If the number of 1's since the last substitution is even, the first and fourth zeroes have a pattern violation.

Another method of achieving 0 synchronisation is called 8-Zero Substitution (B8ZS). Here, a string of 8 zeroes is encoded depending on the polarity of the previous actual 1bit that occurred. If the previous 1 was sent as a negative voltage, the zeroes are sent as three 0's followed by a negative, positive, zero, positive, negative voltage sequence. A similar but opposite pattern violation is performed for the case where thelast 1 was positive. This way a deliberate pattern violation is used to achieve synchronisation.

☞ **Check Your Progress 4**

1) Which type of encoding is most effective at removing the DC component in thesignal and why ?

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2) How does return to zero ensure synchronisation irrespective of the data that istransmitted? What is its disadvantage ?

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3) How can we reconstruct the correct signal inspite of pattern violation in bipolarencoding ?

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3.7 SUMMARY

In this unit, you have seen the need for data encoding and seen that there are four types of data encoding. These arise from the fact that there are 2 source signal types and 2 transmission types. You have seen the different kinds of techniques used for each type of data encoding – analog-to-analog, analog to digital, digital to analog and digital-to-digital. The bandwidth needed, noise tolerance, synchronisation methods and other features of each different technique of encoding have been elaborated upon.

3.8 SOLUTIONS/ANSWERS

Check Your Progress 1

1) Please see the text (Refer Section 3.3, 3.4, 3.5)

2) The spectrum allocation for FM is 88 to 108 MHz, which is 20 MHz. Since abandwidth of 200 KHz is needed for each station, we can have 100 stations in an area. But as discussed, only alternate stations are allowed in one particulararea, to prevent any possibility of interference between them. This means that in any on geographical area, we can have at most 50 radio stations.

3) Please see the text (Refer Section 3.5)

Check Your Progress 2

1) Please see the text.

2) From Nyqist’s theorem, for reconstructing a 1 KHz, signal, we must

sample atleast 2000 times a second, corresponding to a sampling interval of 500 microseconds. However, to be safe and to take care of noise and other perturbations, we may like to sample at a somewhat more frequent interval, such as 400 microseconds.

3) The voltage interval is 12.8v, that has to be quantised into 64 levels. Each level will therefore be 0.2v. Starting from 0v, we find that –3.6v will correspond to alevel of –18, while 0.88v will correspond to a level of +4.

Check Your Progress 3

1) Please see the text.

2) A voltage level of 0 can become anything from –0.2v to 0.2v because of noise. The next detectable level will be 0.7v which can become anything from 0.5v to 0.9v because of noise but will still remain detectable as a separate level. Similarly the next levels that will always remain detectable are 1.4v, 2.1v, 2.8vand 3.5v, making for a total of 10 levels from –3.6v to +3.6v.

3) We can have 18 levels as there are a total of 360 degrees available for shifting.

4) With 10 amplitude levels and 18 phase levels we can have 180 QAM combinations, of which we can then use only 90 for safety.

Check Your Progress 4

1) Bipolar encoding eliminates the DC component entirely because each 0 is represented by a 0 voltage while, a 1 is represented alternately by a positive and a negative voltage. So each 1 cancels out the DC component of the voltage introduced by the previous 1.

2) Please see the text.

3) Please see the text.

3.9 FURTHER READINGS

This unit has tried to cover a vast area in the very little space available. For a more detailed treatment of data encoding, you may refer to the following books:

1) *Computer Communications and Networking Technologies*, by Michael A Galloand William M Hancock, Thomson Asia, Second Reprint, 2002.

2) *Introduction to Data Communications and Networking,* by Behrouz Forouzan,Tata McGraw-Hill, 1999.

3) *Networks,* Tirothy S. Ramteke, Second Edition, Pearson Education, New Delhi, 2004.

UNIT 4 MULTIPLEXING AND SWITCHING

Structure **Page Nos.**

4.0 INTRODUCTION

The transmission of data over a network, whether wireless or not, requires us to solve the problem of being able to send a large amount of data among different groups of recipients at the same time without the message of one being mixed up with those of the others. For example, think of the thousands of radio or television channels that are being broadcast throughout the world simultaneously, or of the millions of telephone conversations that are taking place every minute of the day. This feat is achieved by *multiplexing* the transmissions in some way. In this unit, we will look at the different methods of multiplexing that are commonly used, frequency division multiplexing and time division multiplexing. Time division multiplexing again can be synchronous or statistical.

There are some differences in the characteristics of voice and data communication. Although, finally everything is really data, these differences dictate the design of the telephone network as well as the techniques of switching. When one wants to transmit data over a telephone line, we have to make some changes to be able to get sufficiently high speeds. We will look at ADSL, a scheme for doing this at rates that can compete on cost and quality with those offered by cable television service providers. We will also look at the advantages and disadvantages of the two methods and of high-speed data access.

When constructing networks of any size, a fundamental requirement is that any two nodes of the network should be able to communicate with each other, for, if they cannot do so, they are not on the same network. This brings up the issue of how to switch the data stream, that is, to make sure it reaches its destination and not some other random location on the network. This unit will, describe the different switching techniques available to make this possible.

4.1 OBJECTIVES

After going through this unit, you should be able to:

- state what multiplexing and switching mean;
- state how a telephone line can be used for transmitting data using ADSL;
- describe the differences between ADSL and cable television for data access;

- describe the different kinds of multiplexing and switching;
- state the characteristics of Frequency Division Multiplexing;
- describe the features of Time Division Multiplexing, both synchronous and statistical, and
- differentiate between the different kinds of switching.

4.2 MULTIPLEXING

If, one wanted to send data between a single source and a single destination, things would be comparatively easy. All it would need is a single channel between the two nodes, of a capacity sufficient to handle the rate of transmission based on Nyqist's theorem and other practical considerations. Granted that there would be no other claimants for the resources, there would be no need for sharing and no contention.

The transmission channel would be available to the sole users in the world at all times, and any frequency which they chose could be theirs. In a broadcasting case, we could have the single source transmitting at any time of its choosing and at any agreed upon frequency.

However, things are not so straightforward in the real world. We have a large number of nodes, each of which may be transmitting or receiving, from or to possibly different nodes each time. These transmissions could be happening at the same moment. So, when we need to transmit data on a large scale, we run into the problem of how to do this simultaneously, because, there can be many different sources and the intended recipients for each source are often different. The transmission resource, that is the available frequencies, is scarce compared to the large number of users. So, there will have to share the resource and consequently, the issue of preventing interference between them will arise.

If, all possible pairs of nodes could be completely connected by channels(?!), we would still not have a problem. But that is obviously out of the question, given the very large number of possible source and destination nodes. There would be over a billion telephones in the world today, for instance.

This problem is solved by performing what is called *multiplexing*. It can be done by sharing the available frequency band or by dividing up the time between the different transmissions. The former is called Frequency Division Multiplexing (FDM) while the other scheme is Time Division Multiplexing (TDM).

Another reason for multiplexing is that it is more economical to transmit data at a higher rate as the relative cost of the equipment is lower. But at the same time, most applications do not require the high data rates that are technologically possible. This is an added inducement to multiplex the data so that, the effective cost can be brought down further by sharing across many different, independent transmissions.

In this context, a multiplexer is a device that can accept *n* different inputs and send out 1 single output. This output can be transmitted over a link or medium to its destination, where to be useful, the original inputs have to be recovered. This is done by a demultiplexer. We need to realise that for this kind of scheme to work, the creation of the composite signal by the multiplexer needs to be such that, the original component signals can be separated at the receiving end – otherwise we would end up with just a lot of noise!

When we transmit data over a cable, we are really setting up a different medium. We can transmit data over different cables at the same frequency at the same time without interference because the media or links are different. Even in radio transmission, where the medium is space and hence is a single medium available to all transmissions, geographical distance can in many cases give rise to different links. A low power medium wave transmission happening in India can be done simultaneously with a similar transmission in Europe without interference or the need for any special precautions. However, throughout the rest of the unit, when we talk of multiplexing, we are referring to the simultaneous transmission of data over the same medium or link.

4.2.1 Frequency Division Multiplexing

Suppose, it is human voice that has to be transmitted, over a telephone. This has frequencies that are mostly within the range of 300 Hz to 3400 Hz. We can modulate this on a bearer or carrier channel, such as one at 300 kHz. Another transmission that has to be made can be modulated to a different frequency, such as, 304 kHz, and yet another transmission could be made simultaneously at 308 kHz. We are thus, dividing up the channel from 300 kHz up to 312 kHz into different frequencies for sending data. This is Frequency Division Multiplexing (FDM) because all the different transmissions are happening at the same time – it is only the frequencies that are divided up.

The composite signal to be transmitted over the medium of our choice is obtained by summing up the different signals to be multiplexed (*Figure1*). The transmission is received at the other end, the destination and there, it has to be separated into its original components, by demultiplexing. In practice, a scheme like this could result in interference or cross talk between adjacent channels because the bandpass filters that are used to constrain the original data between the agreed upon frequencies (300 to 3400 kHz) are not sharp. To minimise this, there are guard bands, or unused portions of the spectrum between every two channels.

Another possible cause of interference could arise because of the fact that the equipment, such as amplifiers used to increase the strength of the signal, may not behave linearly over the entire set of frequencies that we seek to transmit. Then, the output can contain frequencies that are the sum or difference of the frequencies used by the input. This produces what is called intermodulation noise.

In this kind of division, it should be realised that the actual modulation technique used is not of consequence. So, one could use analog modulation (AM) or Frequency Modulation (FM). Also the composite signal that we have produced could again be modulated over a different frequency altogether. For example, the three voice channels, that have been modulated for commercial broadcast radio to produce a spectrum from 300 kHz to 312 kHz could be modulated onto a 2 GHz satellite channel for long distance transmission to the other side of the earth. This second modulation could use a technique different from the first one. The only thing that needs to be take care of is, that the recovery of the original signals have to be done in the reverse order, complementing the method used for producing the composite signal.

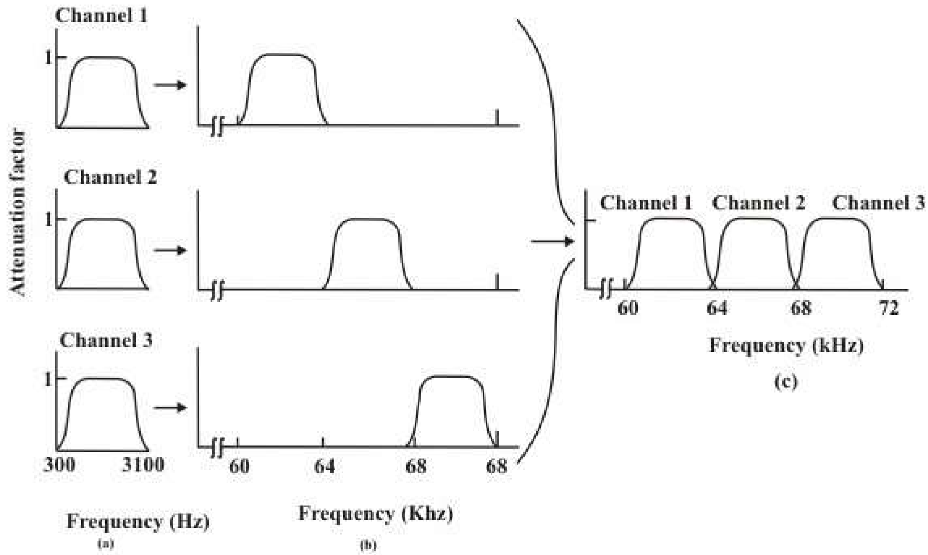

Figure 1: Frequency division multiplexing

[Source: Computer Network by A.S. Tanenbaum]

The International Telecommunication Union (ITU) has standardised a hierarchy of schemes that utilise FDM for the transmission of voice and video signals. To begin with, a cluster of 12 voice channels, each of 4 kHz is combined to produce a signal of bandwidth 48 kHz that is then, modulated and transmitted over the 60 to 108 kHz band. This is called a group.

The next level of the hierarchy is the supergroup, where 5 such groups are combined to use 240 kHz of bandwidth, occupying from 312 to 552 kHz. The next level is the mastergroup that consists of 5 such supergroups. This uses 1232 kHz from 812 to 2044 kHz. Again, 3 such supergroups form a supermaster group that contains 3.872 MHz from 8.516 to 12.388 MHz. The United States uses a similar hierarchy as decided by AT&T (the telecommunications company) that is not entirely identical to the ITU standard.

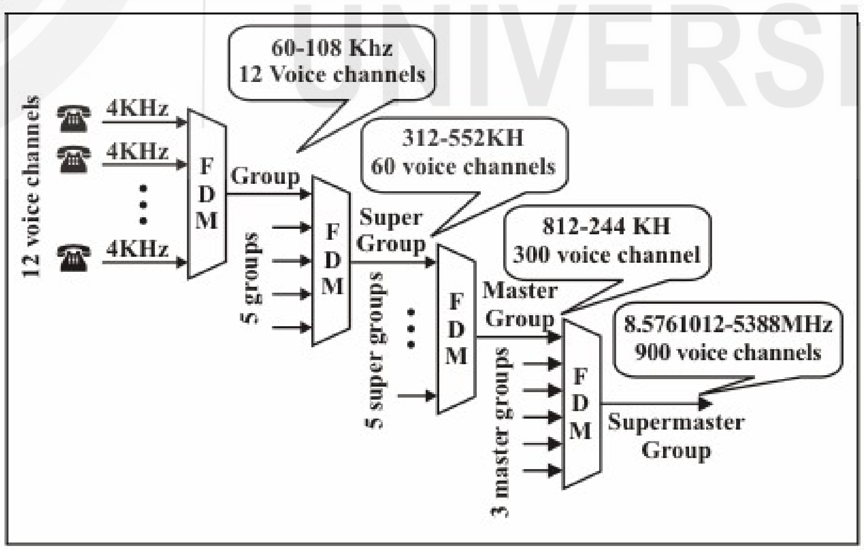

Figure 2: Analog hierarchy

We, thus, see that the original signal could find itself subjected to modulation several times, each of which may be of different kinds, before it is finally transmitted. This raises the possibility of interference and noise, but, with the very good equipment available today, this need not be a matter of concern.

FDM has the disadvantage of not being entirely efficient if the transmissions that are multiplexed together have periods of silence or no data. Since, a frequency band is dedicated to each data source, any such periods are simply not utilised.

4.2.2 Time Division Multiplexing

Another multiplexing scheme is to use the entire bandwidth for each channel but to divide it into different time slots, as shown below.

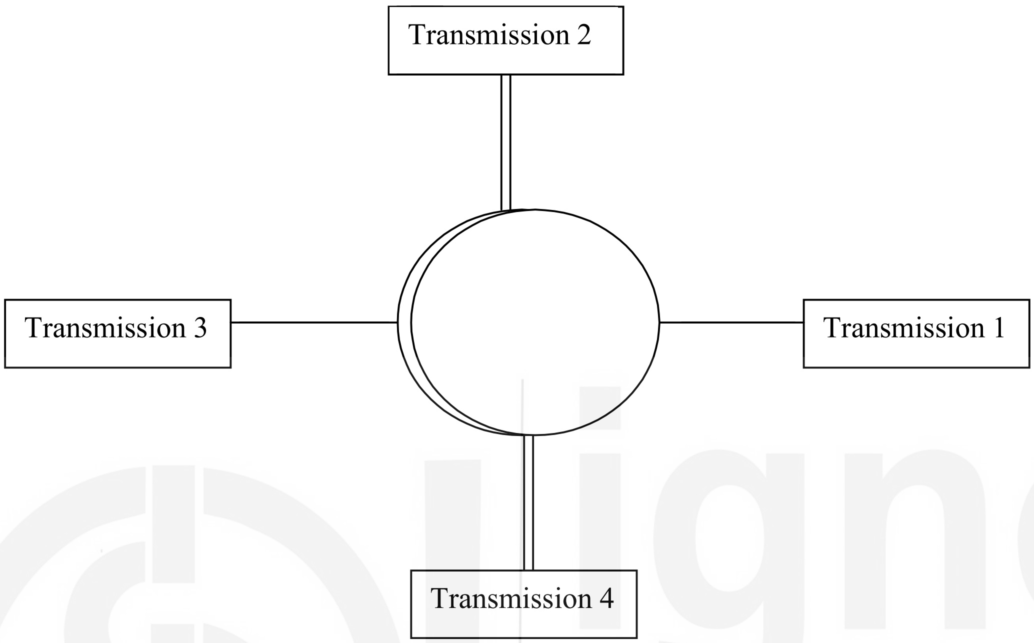

Figure 3: Time division multiplexing

In this case, we have four different transmissions occurring with each being ¼ of the time slice. The slice should be small enough so that the slicing is not apparent to the application. So, for a voice transmission a cycle of 100ms could be sufficient as we would not be able to detect the fact that there are delays. In that case, each transmission could be allotted a slice of 25ms.

At the receiving end, the transmission has to be reconstructed by dividing up the cycle into the different slices, taking into account the transmission delays. This synchronisation is essential, for if the transmission delay is, say 40ms, then the first transmission would start reaching 40 ms later, and would extend to 65ms. If, we interpreted it to be from 0 to 25 ms, there would be complete loss of the original transmission. This interleaving of the signal could be at any level starting from that of a bit or a byte or bigger.

Synchronous Time Division Multiplexing

In this kind of Time Division Multiplexing (TDM), the simpler situation is where the time slots are reserved for each transmission, irrespective of whether it has any data to transmit or not. Therefore, this method can be inefficient because many time slots may have only silence. But it is a simpler method to implement because the act of multiplexing and demultiplexing is easier. This kind of TDM is known as Synchronous TDM.
Here, we usually transmit digital signals, although the actual transmission may be digital or analog. In the latter case, the composite signal has to be converted into analog data by passing it through a modem. Here, the data rate that the link can support has to at least equal the sum of the data rates required for each transmission.

So, if one has to transmit four streams of data at 2400 bps each, we would need a link that can support at least 9600 bps. For each data stream, we would typically have a small, say 1 character buffer, that would take care of any data flow issues from the source.

Although, the transmission has to be synchronous, that is not the reason the scheme is called Synchronous TDM. It is because of the fact that each data source is given fixed slots or time slices. We can also support data sources with differing data rates. This could be done by assigning fewer slots to slower sources and more slots to the faster ones. Here, a complete cycle of time slots is called a frame. At the receiving end, the frame is decomposed into the constituent data streams by sending the data in each time slot to the appropriate buffer. The sequence of time slots allotted to a single data source makes up a transmission channel.

We have already seen that the transmission in TDM must be synchronous because otherwise, all the data would be garbled and lost. How is this achieved? Because this unit is concerned with the physical layer and not the data link layer, we will not concern ourselves with the problem of flow control or error correction or control. But even at the physical layer, we have to ensure that each frame is synchronised to maintain the integrity of the transmission.

One scheme to attain this is to add an extra bit of data to each frame. In succeeding frames, this extra bit forms a fixed pattern that is highly unlikely to occur naturally in the data. The receiver can then use this fact to synchronise the frames. Suppose each frame has n bits. Starting from anywhere, the receiver compares bit number 1, n+1, 2n+1 and so on. If the fixed data pattern is found, the frames are synchronised. Otherwise, the receiver commences to check bit number 2, n+2, 2n+2, … until it can determine whether the frames are in synchronization or otherwise. This continues till, it is able to synchronise the frames and can then start receiving the data from the various channels.

Even after this, it is important to continue to monitor the fixed pattern to make sure that the synchronisation remains intact. If it is lost, the routine of re-establishing it needs to be repeated as before.

The second problem in synchronising the transmission is the fact that there can be some variations between the different clock pulses from the different input streams. To take care of this, we can use the technique of pulse stuffing. Here, the multiplexed signal is of a rate that is a bit higher than that of the sum of the individual inputs. The input data is stuffed with extra pulses as appropriate by the multiplexer at fixed locations in the frame. This is needed so that the demultiplexer at the other, receiving, end can identify and remove these extra pulses. All the input data streams are thus synchronised with the single local, multiplexer clock.

Statistical Time Division Multiplexing

We have seen that synchronous TDM can be quite wasteful. For example, in a voice transmission, much of the bandwidth can be wasted because there are typically many periods of silence in a human conversation. Another example, could be, that of a time sharing system where many terminals are connected to a computer through the network. Here too, many of the time slots will not be used because there is no activity at the terminal. In spite of being comparatively simpler, synchronous TDM is therefore, often not an attractive option.

To take care of this problem, we can use statistical TDM. This is a method where there are more devices than the number of time slots available. Each input data stream has a buffer associated with it. The multiplexer looks at the buffer of each device that provides the input data and checks to see if it has enough to fill a frame. If, there are enough, the multiplexer sends the frame. Otherwise, it goes to the next device in line and checks its input buffer. This cycle is continued indefinitely. At the receiving end the demultiplexer decomposes the signal into the different data streams and sends them to the corresponding output line.

The multiplexer data rate is not higher than that of the sum of all the input devices connected to it. So statistical, TDM can make do with a lower transmission rate than needed by synchronous TDM, at the cost of more complexity. The other way in which this is of benefit is, by the ability to support higher throughput for the same available data rate of the multiplexer. This capability is easily realised during the normal, expected data transmission periods when the amount of data that the different input devices have available for transmission is, in fact, lower than the capacity of the multiplexing device. But the same attribute becomes a disadvantage during peak loads, where all or many of the devices may have data to transmit for a short time. In such a situation, the slack that was available to us for more efficient transmission is no longer present. We will see later how to handle peak load situations in statistical TDM.

In this kind of scheme it is not known to us which input device will have data to send at a given time. So we cannot use a round robin positional scheme that was possible in synchronous TDM. Each frame has to be accompanied by information that will tell the demultiplexer which input device the frame belongs to, so that it can be delivered to the appropriate destination. This is the overhead of statistical TDM, besides increased complexity of equipment.

If, we are sending input data from only one source at a time, the structure of a frame would need to have an address field for the source followed by the data for it. This is really the statistical TDM data frame. It would form the data part of a larger, enclosing HDLC frame if we are using HDLC as the transmission protocol. This frame would itself have various other fields that we will not discuss here.

Such a scheme is also not as efficient as we can make it. This is because the quantum of data available from the source, in that time slot, may not be enough to fill the TDM sub frame. So, while it may be an adequate method if the load is not heavy, we also need to think of a method that can utilise available resources better.

The way to do this would be, to have more than one input data source transmit in a single TDM frame. We would then need to have a more complex structure for the frame whereby, we would have to specify the different input devices in the frame followed by the length of the data field. More sophisticated approaches could be used, in order to optimise the number of bits, we need to encode all this addressingand data information.

For peak loads, there is need for some kind of buffering mechanism, so that, whenever there is excess input from the data sources that the multiplexer cannot immediately handle, it is stored until it can be sent. The size of the buffer needed will increase as the transmission capacity of the multiplexer decreases, as it will become more likely that an input data stream will not be transmitted immediately. It will also depend on the average data rate of the input devices taken together. No matter what the buffer size we choose, there is always a non-zero probability that the buffer itself will overflow, leading to loss of data. If, the average data rate of all devices is close to the peak rate, then we are approaching a situation of synchronous TDM where we are not able to take advantage of periods of silence that is the basis of statistical TDM.

4.3 DIGITAL SUBSCRIBER LINES

Digital Subscriber Lines are an example of multiplexing in action. They are really telephone lines that can support much higher data rates than are possible with an ordinary telephone connection. The concept came up in response to the high data rates of more than 10 Mbps offered by cable television providers. Compared to those rates, the 56 Kbps possible over a telephone dial up connection was a miserable offering.

The 56 Kbps rate is really an artificial barrier arising from the fact that voice telephone line standards limited the bandwidth to about 3.1 KHz to take advantage of the normal range of the human voice in conversation. Nyquist's theorem and the limitations of line quality ensured that we ended up with that rate. Once the 3.1 KHz limit is removed, the telephone line can support much higher data speeds.

Out of the several different technical solutions that came about, the ADSL (Asymmetric DSL) technology has proved to be the most popular. Asymmetric comes from the fact that the data rates in the two directions are different. The approach is to divide the 1.1 MHz bandwidth available over the Cat-3 telephone cables into 256 channels. Of these channels, 0 is used for normal voice communication. The next 5 channels are not used to ensure separation and non-interference between data and voice transmissions. The next 2 channels are used for upstream and downstream control. The remaining 248 channels are available for data transmission.

Like in voice circuits, it would have been possible to use half the channels for communication in each direction. But, statistics show that most users download much more data than they upload. So usually 32 channels are dedicated to uploading, that is, transferring data from the users to the provider and the remaining 216 channels are used for downloading data to the users. This typically translates into 512 Kbps to 1 Mbps download and 64 Kbps to 256 Kbps upload data rates. This then, is the asymmetric aspect of the DSL line.

A problem with ADSL is that the physics of the local loop is such that, the speed at which it can be driven depends heavily on the distance between the subscriber's premises and the provider's nearest termination point. The speed falls sharply with distance and so, distance can become a limiting factor in being able to offer competitive speed compared to that of the cable television providers.

However, ADSL is comparatively simple for a service provider to offer, given an existing telephone network, and does not require much change to its already available equipment. It necessitates two modifications, one each, at the subscriber end and at the end office. On the user's premises, a Network Interface Device has to be installed that incorporates a filter. This is called a splitter and it sends the non-voice portion of the signal to an ADSL modem. The signal from the computer has to be sent to the ADSL modem at high speed, usually done these days by connecting them over a USB port.

At the provider's end office, the signal from the users is recovered and converted into packets that are then sent to the Internet Service Provider, which may be the telephone company itself.

4.4 ADSL Vs. CABLE

At first glance, a comparison between ADSL and cable may seem like a no contest. Cable television, sent over coaxial cables, has a bandwidth that is potentially hundreds of times that of the twisted pair Cat-3 cable used for telephone connections. But, as we go along, it turns out that there are considerations in favour of both sides.

First, there are specific assurances regarding bandwidth that we get from telephone companies who provide ADSL connectivity. An ADSL link is a dedicated connection that is always available to the user, unlike television cable that is shared by scores or even hundreds of subscribers in the immediate neighbourhood. So the kind of speeds that we can get over cable can vary from one moment to the next, depending on the number of users that are working at the time.

There are security risks associated with the fact that cable is shared. Potentially other users can always tap in and read (even change) what you are sending or receiving. The problem does not exist on ADSL, because, each channel is separate and dedicated to the specific user. Though, cable traffic is usually encrypted by the provider, this situation is worse than ADSL where other users just do not get your traffic at all. Also because the channel is dedicated to you, the total number of users does not have any effect on your access speeds as there is no contention with other users.

4.5 SWITCHING

There are many potential sources of data in the world and likewise, many potential recipients. Just think of the number of people who may like to reach one another over the telephone. How can one ensure that every data source is able to connect to the recipient? Clearly one cannot have a physical link between every pair of devices that might want to communicate! Therefore, we need a mechanism, to be able to connect together devices that, need to transfer data between them. This is the problem of switching.

There are two basic approaches to switching. In circuit switching, we create a circuit or link between devices, for the duration of time for which they wish to communicate. This requires a mechanism for, the initiating device to choose the device that it wants to send data to. All devices must also have an identifying, unique address that can be used to set up the circuit. Then, the transmission occurs, and when it is over, the circuit is dismantled so that the same physical resources can be used for another transmission.

The packet switching approach can be used when we are using datagrams or packets to transmit data over the channel. Each datagram is a self-contained unit that includes within itself the needed addressing information. The datagrams can arrive at the destination by any route and may not come in sequence.

4.5.1 Circuit Switching

As mentioned already, in circuit switching, we create a link or circuit between the devices that need to communicate, for the duration of the transmission only. It entails setting up the circuit, doing the actual transmission that can be simplex, half duplex or full duplex, and then dismantling the circuit for use by another pair of devices. An example is shown in the *Figure 4* where there are 7 devices. These are divided into two groups of 3 on the left and 4 on the right. To ensure complete connectivity between them at all times, we would need 12 physical links. But, if connectivity is not required at all times, we can achieve connectivity between any of the devices by grouping them together and using switches to achieve temporary links.

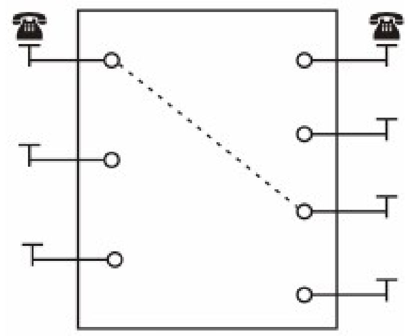

Figure 4: Circuit switching

For example, we can have 7 links that connect the devices A, B and C on the left to the switch. The other devices D, E, F and G on the right are also connected to the same switch. The switch can connect any two devices together using only these seven links as desired.

The capacity of the switch is determined by the number of circuits that it can support at any given time. In the above example, we have seen a switch with 1 input and 1 output. If devices C and E are communicating with each other, the others cannot communicate at the same time, although, there are available links from them to the switch. A circuit switch is really a device that has n inputs and m outputs (n need not be equal to m).

There are two main approaches to circuit switching, called space division or time division switches. Space division switches are so called because the possible circuit paths are separated from one another in space. The old, and now obsolete, crossbar telephone exchanges are an example of space division switching. The technique can be used for both digital or analog systems. There were other designs of such switching but the only one that went into large scale use was the crossbar. Because of the way it is constructed, such switching does not have any delays. Once the path is established, transmission occurs continuously at the rate that the channel can support.

In essence, a crossbar connects p inputs to q outputs. Each such connection is actually performed by connecting the input to the output using some switching technology such as, a transistor based electronic switch or an electromechanical relay. This requires a large number of possible connections, called crosspoints. For a 10,000 line telephone exchange, it would mean that, each of the 10,000 possible inputs be able to connect to any of the 10,000 possible outputs, requiring a total of 100,000,000 crosspoints. As this is clearly impractical, the pure crossbar design is not usable on a commercial scale. Moreover, there are inherent inefficiencies in this design as statistically, only a small fraction of these crosspoints are ever in use simultaneously.

The way to get around this limitation is, to split the switch into different stages. If we consider a 36 line exchange where each of 36 inputs needs to be connected to 36 outputs, we can do so in, say, 3 stages. The first stage could have 3 switches, each with 12 inputs and 2 outputs to the two second stage switches. These intermediate switches could each have 3 inputs from the 3 first stage switches and 3 outputs to the 3 third stage switches. The last stage of the switches would then have 2 inputs from the 2 second stage switches and 12 outputs, each to 12 of the 36 devices. It is thus, possible for each of the 36 inputs to connect to each of the 36 outputs as required, using only 72 + 18 + 72 = 162 crosspoints, instead of the 1296 crosspoints that would have been required without the multistage design. The following *Figure 5* shows the multistage switch.

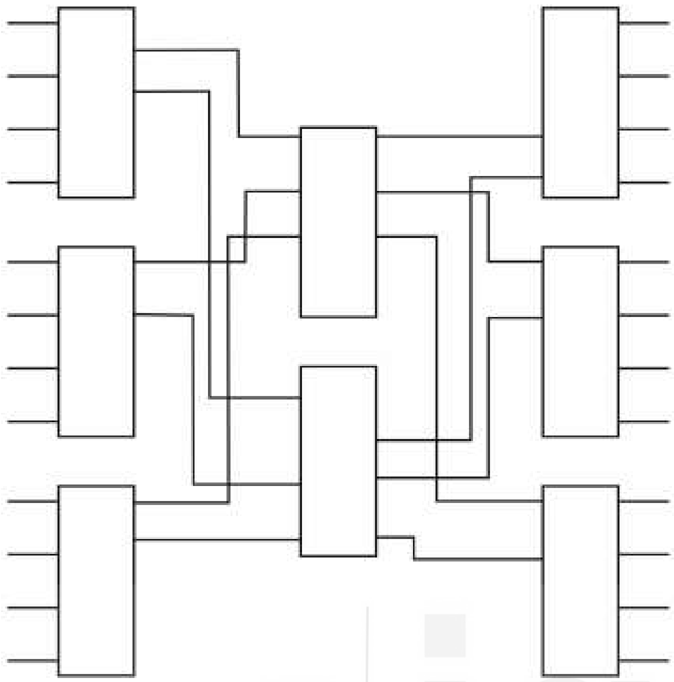

Figure 5: Multistage switch

If, we use a single stage switch, we will always be able to connect each device to any other device that is not already busy. This is because all the paths are independent and do not overlap. What this means is that, we will never be starved of circuits and will never suffer the problem of blocking. For multistage switching, the reduction in the number of crosspoints required comes at the cost of possible blocking. This happens when there is no available path from a device, to another free device, that wedesire to connect to. In the *Figure 5*, if we have a switch in the first stage that is already serving 2 input devices, then there are no free outputs from that stage. We cannot therefore service more than 2 input devices connected to one switch at a time and the 3rd device would not be able to connect anywhere, getting a busy signal.

It is possible to minimise the possibility of blocking by using stochastic analysis to find the least number of independent paths that we should provide for. However, we will not go into such an analysis in this unit. But, we can see that the limiting factor in the *Figure 5* is the middle stage where there are only 3 inputs and 2 outputs. This is the point at which congestion is most likely to occur. You too may have experienced blocking when trying to make a telephone call and found that you got an exchange busy tone, indicating that it was not the number you called but the exchange (switch) itself that was congested. Such a problem is, most likely to occur during periods of heavy activity such as at midnight on a new year's day, when many people might be trying to call one another to exchange greetings.

Another advantage of multistage switching is that, there are many paths that can be used to connect the input and output devices. So, in the above case, if there is a failure in one of the connections in the first stage switch, two input devices can still connect to it. In the case of single stage switching, that would have meant that we could not set up a circuit between those devices.

Let us, now look at another method of switching that uses time slots rather than spatial separation. You have already seen how synchronous TDM involves transmission between input and output devices using fixed time slots dedicated to each channel. But, that is not switching because the input-output device combinations are fixed. So, if we have three devices A, B and C transmitting and three devices D, E and F that are receiving the respective transmissions, there will be no way to change the circuit path so that A can transmit to E or F.

To achieve switching, we use a device called a Time Slot Interchange (TSI). The principle of such a device is simple. Based on the desired paths that we want to set up, the TSI changes the input ordering in the data streams. So, if the demultiplexer is sending the outputs to D, E and F in order, and if we want that A send data to E instead of F, and that B send data to F rather than E, then the TSI will change the input ordering of the time slots from A, B, C to C, A, B. The demultiplexer will not be aware of this and will continue to send the output the same way as before. The result will be, that our desired switching will be accomplished. However, unlike space division switching, our output will be subject to delays because we might have to wait for the input from the right device before it can be transmitted. This is unlike space division switching where the data can be sent in a steady stream.

How does the TSI work? It consists of a control unit that does the actual reordering of the input. For this, it has to first, buffer the input it gets, in the order it gets it. This would be stored in some kind of volatile memory. The control unit then sends out the data from the buffer in the order in which it is desired. Usually the size of each buffer would be that of the data that the input generates in one time slice.

We do not have to confine ourselves to a single type of switch. There can be switches that are based on a combination of both kinds of switching. For example, we could have a multistage switch where, some of the stages are space division switches whiles others are time division switches. With such an approach, we can try to optimise the design by reducing the need for crosspoints while keeping the delays in the whole system to a minimum. For example, we could have a TST three stage switch where the first and last stages use time division switching while the middle stage uses crosspoints.

Circuit switching is useful for voice based communication because of the characteristics of the data transfer. Although, a voice conversation tends to have periods of silence in between, those periods are usually brief. The rest of the time there is data available for transmission. Secondly, in such communication, we cannot tolerate delays of more than about 100 ms as that becomes perceptible to the human ear and is quite annoying to the speaker and listener.

Again, because it is human beings that are present at both ends, the rate at which data is generated at both ends is similar, even if one person talks a bit faster than the other! Also, the other human can usually understand what is being said even if he cannot at the same rate. And if really required, one can communicate to the other to speak slower or louder. But, when we are dealing with data generating devices, there can always be a mismatch between the rate at which one device generates data and at which the other device can assimilate it. Moreover, there can be long periods when there is no data generated at all for transmission. In such a situation, circuit switching will not be a suitable method and we have to look at something that takes care of the characteristics of data communication between devices.

4.5.2 Packet Switching

Packet switching *Figure 6* is a method of addressing these problems of circuit switching. It will be further elaborated in the Block 3. It is based on the concept of data packets, called datagrams. These are self contained units that include the address of the destination, the actual data and other control information. It takes care of the sporadic nature of data communication where transmissions tend to occur in bursts. So, we do not waste transmission capacity by keeping circuits connected but, idle while there are periods of silence. Even if we multiplex the channel, we cannot cater to a situation where all or most of the devices are silent, leading to underutilisation of the capacity. The concept is so interesting that it is worth devoting few words to it. Whenever a user wants to sends a packet to another user, s/he transmit to the nearest router either on its own LAN or over a point-to-point link to the caviar. The packet is stored for verification and then transmitted further to the next router along in way until it reaches for fixed destination machine. This mechanism is called packet switching.

Some of the other limitations of circuit switching for sending data are:

- It is not possible to prioritise a transmission. In a circuit switching mechanism, all the data will be sent in the order in which it is generated, irrespective of its importance or urgency. This is fine for voice transmission, in which we want to hear things in the sequence in which they are spoken at the other end, but this does not work for data where, some packets may be more important than others and the order of delivery does not have to be sequential.

- A circuit, once set up, defines the route that will be taken by the data until it is dismantled and set up again. Sometimes, that circuit may have been set up via a less advantageous set of links because that was the best route available at the time it was set up (best could be in terms of channel capacity, delays, line quality or other parameters). Now, subsequently, even if another, better route is released by other devices, we cannot change over to this better route without disconnecting the previous circuit and forcing the participants to set up the call again.

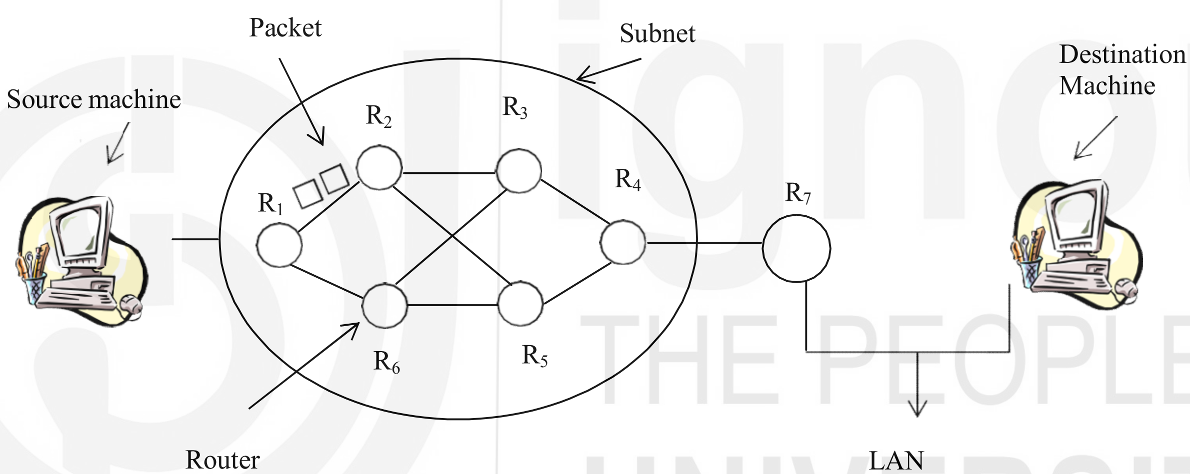

Figure 6: Packet switching

A datagram can contain different lengths of data. Besides, it would have control information such as the address to which it needs to be sent, the address from which it has originated, its type and priority and so on. From its source node it is sent to a neighbouring node, depending on its final destination. The format of the datagram depends on the protocol that is used and it is the responsibility of each node to route the datagram onto the next node that would take it to its destination. There can be several routes possible and each node would make the decision based on availability, traffic conditions, channel quality and so on.

Although datagram length can vary, long messages would have to be split into more than one datagram as the network protocol would have a limit on the maximum size of a datagram. Because each datagram is routed independently, the different datagrams that make up a message may arrive out of sequence at the destination, as different routes may take different times to traverse. In communication protocols it is the responsibility of the network layer to make sure that routing gets done, while it is the transport layer that ensures that the message at the receiving end is in the proper sequence.
Again, the physical link between two nodes can be of any type and is of no consequence at the higher layers of the communication protocol in use. That link itself may be multiplexed and may carry several transmissions simultaneously, for the

same or different pairs of source and destination nodes. Moreover, these transmissions may be happening in different directions.

This was the datagram approach to packet switching. It is also possible to look at packet switching in terms of a virtual circuit, which again has two different approaches that are in current practice.

In a virtual circuit, when the first datagram is sent out, we decide on the route that will be followed, and subsequent datagrams continue to follow that route. So, it is like circuit switching to a large extent. But, in circuit switching the link is dedicated to the pair of nodes and there is no multiplexing that happens at the level of individual switches. In virtual circuits there can be multiplexing at the switches as well.

Permanent virtual circuits are dedicated channels set up for communication between a pre-decided pair of nodes. It is thus, akin to a leased line where the routing is decidedonce the virtual circuit is set up. There is no call set up required because, the link is always available to the pair of nodes. On the other hand, we can also have switched virtual circuits where the routing is decided every time a pair of nodes want to communicate. This is like a dial-up connection where any two pairs of nodes can communicate by setting up a call that is terminated when the conversation is over.

Another difference between circuit and virtual circuit switching has to do with reliability and failure recovery. In circuit switching, if the link breaks because of a failure in any portion, the call is terminated. It would have to be established again forthe conversation to continue. In virtual circuits, although the route is decided at the beginning when the session is set up, in case of failure of any part of the route, an alternate route will be agreed upon and set up.

Also, in circuit switching there is no possibility of failure due to congestion after the link is set up. In virtual circuits, congestion can occur even later because of multiplexing at the switches.

☞ Check Your Progress 1

1) What is the problem in offering high speed ADSL connection to all subscribers that have a telephone?

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2) Which method is more efficient in terms of capacity utilisation?

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3) Do you think voice transmission could be done using packet switching?

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4) What sort of problems could arise in trying to completely connect all data transmissiondevices in the world?

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5) What are some of the requirements for a multiplexed signal to be useful?

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6) Is multiplexing needed for transmissions over different links?

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7) What are the problems that can occur in Frequency Division Multiplexing?

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8) What is the method that FDM uses to mix signals that can be recovered later?

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9) List the considerations that make FDM useful and possible.

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10) What are the considerations in choosing the length of the time slice for Time Division Multiplexing?

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11) Why must the multiplexer and demultiplexer be synchronised? Why then is synchronous TDM so called?
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12) What are the problems in synchronising the transmission and how can they be taken care of?
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13) What are the inefficiencies inherent in synchronous Time Division Multiplexing and how does statistical TDM seek to reduce them?
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14) What price do we have to pay for increased efficiency in statistical TDM?
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15) What would happen if the load from the input devices exceeds the capacity of the multiplexer?
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16) How does ADSL enable high speed data access although voice lines are so slow?
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17) Why is ADSL called asymmetric and why is it not kept symmetric?
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18) How can a cable provider improve quality of service after the number of users becomes larger?
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19) Why is ADSL more secure than cable for data communication?
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20) Why is switching necessary?
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21) What is circuit switching?
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22) What is packet switching? How does it differ from circuit switching?
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23) What is meant by multistage switching? Is the capacity of such a switch limited?
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24) List five important features of space division switching.
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25) What are the characteristics of time division switching?
……………………………………………………………………………………

26) How can we combine space and time division switching? What are the advantages of such an approach?
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27) Why is circuit switching suitable for voice transmission?
……………………………………………………………………………………

28) What are the features needed for a switching mechanism that transmits data?
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29) What is a virtual circuit? How does it differ from circuit switching?
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30) What are datagrams and how can they be used to transmit data?
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4.6 SUMMARY

Multiplexing is needed in communication networks because of the scarcity of bandwidth compared to the large number of users. Frequency Division and Time Division Multiplexing are the two ways of multiplexing, of which Time Division multiplexing can be synchronous or asynchronous. The asynchronous method is more complex but more efficient for data transmission.

ADSL is a means of utilising the existing capacity of the local loop in the telephone system for providing subscribers with high speed data access. Such access is also

provided by companies over the cable television network. ADSL is more secure and predictable in terms of service quality, while cable does not have the limitations of distance from the end office that ADSL has.

Switching is necessary to connect two nodes or devices over the network that intend to communicate for a limited duration. Circuit switching is more suitable for voice communication, and can be done using space division, time division or a combination of both kinds of switches. Multistage switching is needed to optimise the number of crosspoints needed. Packet switching is used for data transmission and allows for prioritising of data packets, alternate routing as needed and is also more efficient for the bursty traffic pattern of data communication. Datagrams are self contained packets of data that are routed by the intermediate nodes of the network. Switched or permanent virtual circuits can also be utilised, where the route is established at the beginning of the session, but can be altered without disrupting the channel in case of failure of any part of the route.

4.7 SOLUTIONS/ANSWERS

Check Your Progress 1

1) First, the subscriber may be located too far from the provider's end office for the local loop to give sufficiently high data speeds to be competitive with cable. Secondly, providing a connection means a visit to the subscriber's premises to install the Network Interface Device, splitter and ADSL modem.

2) Cable is more efficient because bandwidth is not dedicated to a user. In ADSL, a user's dedicated bandwidth remains unutilised if s/he is not active at any given time.

3) Packet switching can be and is used for voice transmission, such as in the highly popular Voice Over Internet Protocol (VOIP) used for Internet Telephony. It is cheaper than voice transmission though there is some deterioration in voice quality.

4) There are a very large number of such transmission devices. For example, there may be over a billion telephones in the world today. The number of data channels to always connect even one phone to another billion phones alone would be one billion. It is therefore, not practically possible to have so many distinct data channels.

5) Multiplexing is the creation of a composite signal from various constituent signals at the transmitting end. There must be a way to separate the different signals at the receiving end, otherwise all we would have at that end would be a lot of noise. Secondly, there must be a way to send each separate signal to the proper device for which it was meant, otherwise we may have a useful signal but delivered to the wrong device.

6) Multiplexing is needed because we want to send several signals over the same link. If there are different physical links available, then no multiplexing is needed because the transmissions would be independent and would not interfere with each other.

7) One problem with FDM is that we have to use filters to constrain the constituent signals between the agreed upon frequencies. For example, in telephone transmission, the voice signal is filtered to lie between 300 and 3400 Hz. Other frequency components in the voice, such as very low or high frequencies, are therefore not transmitted. This filtering is not very sharp and can result in cross talk between adjacent channels.

   Secondly, the amplifiers and other equipment used may not behave linearly over the whole frequency range. This gives rise to other frequency components that were not present in the original signal, giving rise to distortion of the original signal. This is called intermodulation noise.

8) FDM uses different methods to mix signals. We can use analogue modulation (AM) where the carrier amplitude is modulated with the signal, or frequency modulation (FM) where the carrier frequency is modulated with the signal.

9) FDM is useful because we are able to make use of the available spectrum more efficiently. We can use different carrier frequencies to send different signals at the same time.

10) The length of the time slice must be such that the application that is using the signal is not able to perceive the slicing, or it does not matter to the application. For example, in

voice transmission, a delay of 300 ms is quite apparent to a human speaker. So the time slice must be much smaller than this.

11) The multiplexed signal consists of signals from the different data sources, each of which has a fixed time slot. To send the received data to the correct data source, the demultiplexer must be in synchronization with the multiplexer, else the data would go to the wrong destination. That is why synchronous TDM is so called.

12) One problem is that the clocks of the different input signals may not be in synchronization. So we use a multiplexer clock to do the synchronization. The multiplexed signal is sent at a rate slightly higher than the sum of the input signals. At fixed points in the transmitted data frames, we stuff the frame with synchronization pulses that can be identified and removed by the demultiplexer at the receiving end.

Secondly, the start of the frame from different sources needs to be known. This is done by adding extra bits to the frames that form a fixed pattern unlikely to occur in an actual transmission. This pattern is used by the demultiplexer to establish synchronization with the input from the multiplexer.

13) In synchronous TDM, every data source has a fixed time slot, irrespective of whether ithas any data transmit during that slot or not. For example, a human conversation is often interspersed with periods of silence. Such slots are really wasted because no useful data is sent in that slot. Statistical TDM seeks to improve efficiency by ensuring that every slot is utilized. If a particular data source has no data to send, then the slot is given over to another data source. To make it even more efficient, if a particular data source does not have enough data to utilize the whole time slot, then within one time slot we send data from more than one data source.

14) Statistical TDM is a more complex scheme to implement and requires more complex equipment at both the multiplexing and demultiplexing ends. But this is well worth the increased throughput one gets.

15) Such a situation would cause loss of data. To prevent such loss, we have a buffer in themultiplexer to store such excess data. The size of the buffer depends on how often we expect such a situation to occur and for how long it would persist. If we have a situation where the peak load from the devices is close to the average load, then there islittle advantage to using statistical TDM and a simpler synchronous TDM approach would be adequate. This is because the basic premise of statistical TDM that each data source has periods of silence.

16) Voice lines are not slow because of any inherent limitation. They are kept artificially slow because that is the nature of voice transmission, given the 3400 Hz upper limit on most normal voice conversations. They therefore do not need to cater to higher frequencies and do not need a higher transmission rate. Once we remove these constraints for data transmission, high speed data access is possible over the same lines.

17) ADSL is called asymmetric because the data transmission rate available to the user for uploads and downloads are different. Of the 248 channels available for the transmission, 32 are dedicated to uploads and 216 to downloading. This is done because in practice most users download much more than they upload. So it improves the performance of the line. However, it could have easily been kept symmetric.

18) The option he would have would be to install an additional cable to serve some of the users.

19) ADSL is more secure because each link is physically independent of that used by other users. In cable, all users share the same channel with the potential to tap into the data of other users. However, the problem can be mitigated by encrypting all traffic.

20) Switching is necessary because we cannot completely connect all the devices that may have a need to communicate, as that number is very large. So whenever needed, we have to use some mechanism to connect them together.

21) In circuit switching, a dedicated path or channel is set up between the devices that want to communicate. This requires each device to have an identifying address by means of which the initiating device can request for a channel. For the duration of the conversation, the channel is dedicated to the two devices. Once the conversation is over, the circuit is dismantled and the same physical resources can be used for another pair of devices that might want to communicate.

22) In packet switching there is no dedicated channel set up for the two devices to communicate. The transmitting device merely sends out the message over the transmission medium that can be accessed by all devices. The intended recipient finds out that the message is meant for it from the address and retrieves the message.

23) In multistage switching, there are different points at which switching is done to set up the circuit, rather than having a direct path between each device to be connected. The capacity of such a switch is limited by the number of available paths at a time. So the reduction in the number of crosspoints is at the cost of possible blocking.

24) In space division switching
   a. The different paths are separated in space
   b. There are no delays during the transmission
   c. There is a set up time while the circuit is established
   d. During periods of silence, resources are wasted
   e. The number of crosspoints limits the capacity of the switch

25) Here switching is achieved by multiplexing the transmission from different sources anddemultiplexing them at the output stage. There can be delays while the switching is being done. Flow control is also more of a problem here.

26) Different stages of a multistage switch can use different approaches. The time division switches help reduce the number of crosspoints while the space division switches reduce the delays. By using a combination of both one can try to optimize the number of crosspoints without having too much delay.

27) Voice conversations do not have many periods of silence, so that the circuit is used effectively. Humans can communicate with each other and ensure that flow control is done, say, if somebody is talking too fast for the other to understand. Moreover, in voice conversations we cannot tolerate delays and so circuit switching is the better choice as it gives us a dedicated channel.

28) For transmitting data, we need a switching mechanism that allows for
   - Efficiency during periods of silence. In circuit switching, the available resources are wasted during such periods as the link is dedicated to the devices and cannot beused by others.
   - Flow control is required because the sending and receiving devices may not have the capacity to communicate at the same rate. In circuit switching, such a situationwould lead to loss of data.
   - If a route becomes unavailable, we should be able to change over to another route.Similarly if a better route becomes available, we should be able to utilize it.

   All the above features are provided by packet switching.

29) A virtual circuit is set up at the time a pair of devices begins to communicate using packet switching. Unlike circuit switching, we can change the path or route used if there is a failure in the route. Virtual circuits can also be permanent, that is, dedicated to a pair of devices, or can be set up for a limited time between different devices. Another difference is that in virtual circuits multiplexing can happen at the level of individual switches as well.

30) Datagrams are self contained units of information that consist of a control part that has data about the recipient, the priority and so on, as well as the actual data to be sent. Each node that receives a datagram is responsible for sending it out onto the next node till it reaches its destination. Datagrams may arrive out of sequence at the destination because they have taken different routes.

4.8 FURTHER READINGS

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2) *Data and Computer Communication*, William Stalling, PHI, New Delhi.
3) *Introduction to Data Communications and Networking,* by Behrouz Forouzan, Tata McGraw-Hill, 1999, New Delhi.